

Planning and Design of a Drinking Water Reservoir and Dam

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1.0 INTRODUCTION

In the face of growing climate variability and rising demand on water infrastructure, the need for a consistent and sustainable water supply system has become more urgent. [1] The project focuses on the planning and execution of the design of a drinking water reservoir dam, which is intended to improve the water supply in the region of Thuringia, Germany, while also enhancing the flood protection capacity. The project is guided by DIN 19700 standard and based on realistic and geological data provided for the region. [2]

Thuringia, often referred to as the “Green Heart” of Germany, faces a range of challenges, including a long-term dry period, ecological stress on its river systems, and reduced water quality linked with past events, including mining and agricultural activities. As a result of this, the dam design must strike a careful balance between technical feasibility and environmental sustainability. The project is mainly organized into three main parts, which include developing the overall planning concept, which acts as a guide in the other two parts. Part two deals with water management and key calculations such as water storage volume using HQ100 values and design of flood events, BHQ1 and BHQ2. Part three is mainly focused on designing dams, which consists of the definition of storage volume, spillway design, and bottom outlet, checking for stability, and cross-section. [1]

In recent decades, Thuringia has experienced a noticeable increase in hot, dry days, making the necessity of a new dam. The new dam is intended to provide secure drinking water for the region and also support agricultural use. In addition to this, an existing upstream dam “Vor-sperre Deesbach” currently plays a temporary role in water supply and will be considered in the overall concept to ensure compatibility with the new reservoir system.



Figure 1: Map of Thuringia Region [3]

1.1 DATA PROVIDED

The following data were given and are used in the report:

1. Monthly value of drinking water demands for the region.
2. Discharge data from Unterweißbach gauging region covering the year 1969 to 2019 including both mean flow and peak flow.
3. Digital Terrain Model (DTM) covers the catchment and reservoir area.
4. Geological data describing the surface condition and soil composition.
5. Precipitation data from KOSTRA data set.
6. Relevant design standards including DIN 19700 and other technical guidelines.

2.0 STUDY AREA

Thuringia is a region in the heart of Germany, known for its green landscape, deep forest and history. The region is surrounded by other five German states covers about 16,200 square kilometers. [4] One of its famous natural area is the Thuringian Forest (Thüringer Wald), a low mountain range mostly with hiking trails and scenic views. In the center of Thuringian basin, a flat fertile area where farming is common for grains and vegetables. Rivers such as the Saale, Unstrut, and Werra flow through the land, shaping valleys and providing water to both nature and people. [4] These natural features make Thuringia a region with a strong connection to the environment, offering clean air, fresh water, and lots of green space.

The capital city is Erfurt, it is a lively mix of old and new with a preserved, medieval center and modern economy. Other cities like Jena and Weimar play an important role. Jena known for science, optics and technology, while Weimar is a symbol of German culture and history. Gera is another important city with a strong industrial background. While the cities are growing, much of Thuringia remain peaceful and rural with forest, small town and farming village. The region is well connected by roads and trains making it easy to accessible.

2.1 TOPOGRAPHY

The topography of Thuringia presents a diverse and well-balanced landscape, shaped by both natural formations and long-standing land use. [4] The southern part of the region is dominated by the Thuringian Forest, a long, narrow mountain range covered in dense coniferous and deciduous forests. This upland area is characterized by rugged terrain, including steep slopes, forested ridges, and deep valleys. The highlands are ecologically significant, offering habitats for a wide variety of plant and animal species. Additionally, the forested slopes serve as important natural water catchments, where rainfall is absorbed and gradually released into the river systems, helping to regulate flow and reduce erosion. The southern terrain also supports outdoor recreation and tourism, making it both environmentally and economically valuable.

On the other side, the northern and central parts of Thuringia are shaped by the Thuringian Basin, a broad, low-lying area with gently rolling hills and wide, open plains. The smooth transition from the elevated, forested south to the flatter, agriculturally productive north gives Thuringia a topographic variety that benefits both nature and human use. [4] This combination of uplands and lowlands is one of the region's most defining geographical features.

2.2 CLIMATIC CONDITIONS

The climate of Thuringia can be classified as temperate, situated in a transitional zone between maritime and continental influences. [1] The region experiences moderate seasonal variability in both temperature and precipitation, shaped by its central location in Germany and varied topography. According to long-term observations, the average annual air temperature in Thuringia is approximately 9.8 °C, with a corresponding mean annual precipitation of 571 mm. These values reflect the region's overall climatic balance, though significant local differences occur between the uplands of the Thuringian Forest and the lower Thuringian Basin. [1].

Data collected over 17 years at the Jena experiment site provide valuable insights into the weather dynamics of Thuringia. The observations show regular seasonal cycles, as well as variations driven by large-scale air circulation patterns, elevation, and local land cover. The southern highlands, especially the Thuringian Forest, tend to be cooler and wetter, with more frequent snowfall in winter and higher precipitation levels throughout the year. In contrast, the central and northern parts, including the Thuringian basin, experience warmer summers, milder winters, and slightly lower precipitation, making these areas more favorable for agriculture.

Over the past five decades, regional climate trend analyses in Saxony, Thuringia, and Central Germany have shown a clear warming pattern, averaging around 0.2 °C per decade, along with shifting rainfall patterns characterized by a decrease in summer rainfall and a slight increase in winter precipitation. [5] These trends suggest increasing climate variability, with potential implications for agriculture, forestry, and water management.

In comparison with the Elbe and Oder river catchments, which are influenced by drier continental climatic conditions in eastern Germany, highlight important hydrological contrasts. In these regions, reduced precipitation and lower soil moisture lead to slower catchment response, reduced sediment transport, and less frequent flooding compared to basins in more humid climates. These findings underscore how even moderate shifts in climate can significantly affect river system behavior and watershed management.

2.3 POPULATION

As of 31 December 2023, Thuringia's population was officially recorded at 2,114,870 inhabitants, according to estimates from the Federal Statistical Office of Germany. This figure reflects a slight decline from earlier estimates, such as the 2,100,277 projected for 31 December 2024 [6]. Driven primarily by demographic trends including low birth rates, an aging population, and the migration of younger residents to larger urban centers. The population density in the region stands at approximately 130.5 inhabitants per square kilometer.

Thuringia is experiencing a steady population decline caused by both natural factors and changes in its population structure. In 2022, the region recorded about 14,000 births and nearly 27,000 deaths, resulting in a negative natural growth, meaning more people are dying than being born each year [6] [7]. This is a common trend in many rural areas of Germany. One of the key reasons for the low birth numbers is the uneven gender balance in the 25 to 40 age group, which is the typical age range for starting families. In this group, there are approximately 115 to 120 men for every 100 women, mainly due to the outmigration of young women to larger cities for education or work opportunities. This imbalance reduces the number of potential mothers in the region, which continues to put downward pressure on the birth rate and contributes to the long-term decline in population.

3.0 CATCHMENT CHARACTERISTICS

Catchment delineation is a critical step in hydrological analysis, providing the spatial foundation for understanding runoff behavior, water availability, and system design. For the region of Unterweißbach, detailed mapping of the catchment area was carried out using a Digital Terrain Model (DTM) in combination with QGIS software. The evaluation focused on two specific areas: the broader catchment contributing to the gauging station and the smaller sub-catchment relevant to the planned dam location. The analysis determined that the catchment area upstream of the gauging station covers approximately 83.62 km², while the catchment influencing the dam site encompasses about 72.6 km². This difference is essential for understanding which portions of the landscape actively contribute to surface runoff at specific control points. By isolating effective catchments, hydrological models can be calibrated more accurately, supporting effective decision-making for flood control, reservoir operation, and ecological planning.

Catchment Characteristics	Values
Catchment area of gauging station	83.62 km ²
Catchment area (A _E)	72.6 km ²
Ratio	0.86
Length of main river (L)	17.6 km
Length of all rivers ($\sum L$)	93.34 km
River density	1.29
Elevation of gauging station	320
Highest Elevation	850
Height difference (Δh)	530
Average river slope (I)	0.02984

Table 1 : Catchment Characteristics

The catchment area, determined using QGIS software, is shown in the following figure along with additional details

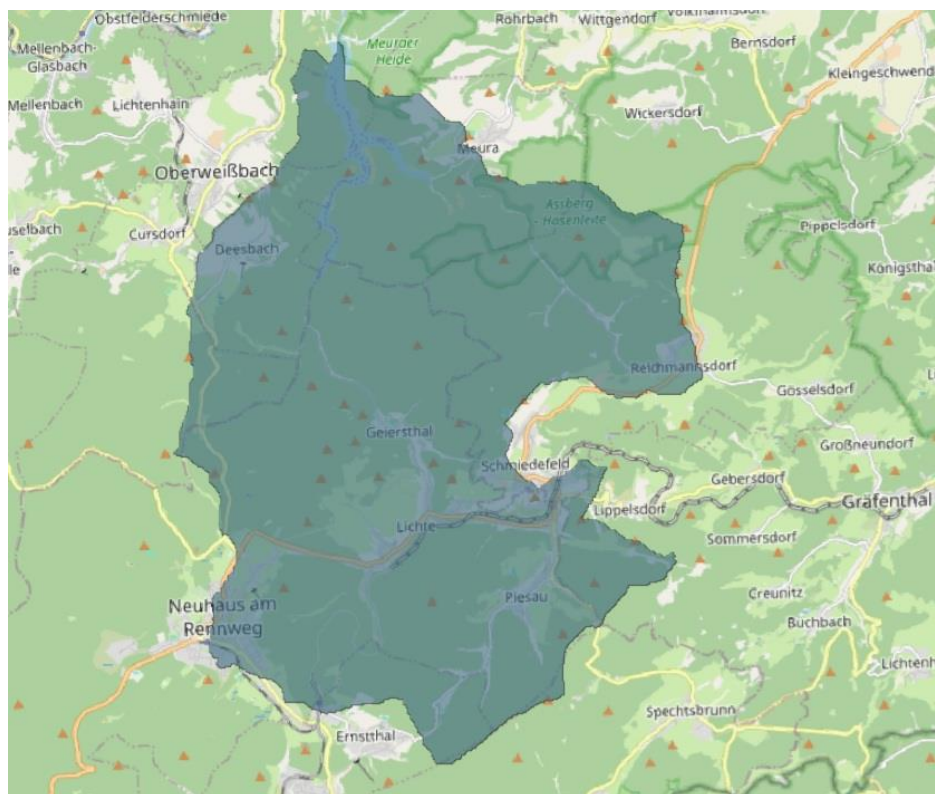


Figure 2: Catchment area determined by QGIS software

4.0 RESERVOIR CAPACITY PLANNING

Planning for reservoirs is crucial to maintaining water supply, ecological equilibrium, and flood resistance. The objective is to determine the amount of water that must be stored and managed throughout the year to meet human needs while also meeting environmental requirements. This requires a deep understanding of regional hydrology and seasonal fluctuations, as well as long-term changes in climate.

Well-designed reservoirs are more than just a storage location for water; they also provide drinking water, hydropower generation, as well as safeguarding against droughts and floods. Keeping all these goals in mind, storage must be divided into functional zones that serve a variety of purposes, from operational reserves to safety nets during harsh weather conditions. The use of data-driven methods and regulatory standards in modern reservoir planning enables the management of water for communities and ecosystems while maintaining efficiency and sustainability.

4.1. DAM TYPE

A Roller-Compacted Concrete (RCC) dam has been chosen as the dam type for the current water resource infrastructure project in Thuringia, Germany. This choice was made considering the region's hydrological, geotechnical, and economic factors. RCC dams are especially well-suited for moderately tall structures in locations where durability, affordability, and speed of construction are valued, all of which fit the project's local specifications.

4.1.1. REGIONAL SUITABILITY AND GEOTECHNICAL CONSIDERATIONS

Thuringia's geological formations, which primarily consist of clayey soils and stable bedrock substrata, are ideal for load distribution and anchoring RCC foundations. Central Germany's seasonally variable hydrological regimes and moderate seismic risk can be effectively adapted to using the RCC method. The use of RCC is further justified by the high groundwater table found in the catchment analysis, since its dense structure reduces seepage and guarantees long-term water tightness.

4.1.2. MATERIAL OPTIMIZATION USING LOCAL RESOURCES

To enhance the concrete mix's workability and thermal performance, the design makes use of locally sourced aggregates and the possible addition of additional cementitious materials, such as fly ash from adjacent thermal plants. In addition to lowering material costs, this helps Germany achieve its carbon reduction targets and emphasize sustainable building methods.

4.1.3. CONSTRUCTION EFFICIENCY AND ECONOMIC ADVANTAGE

The facilitated construction process provided by RCC is a significant benefit in an area such as Thuringia, where project schedules frequently rely on brief summer construction windows. By continuously placing no-slump concrete in thin lifts (usually 30 to 60 cm) and compacting it with vibratory rollers, the technique drastically cuts down on formwork and construction time. This permits early reservoir impoundment and guarantees the timely achievement of infrastructure objectives, including flood control, drinking water supply, and ecological flow maintenance.

4.1.4. STRUCTURAL AND OPERATIONAL RELIABILITY

With its solid design, the RCC dam offers high resistance to hydraulic uplift pressures, dynamic loads, and freeze-thaw cycles, all of which are crucial in areas like Thuringia that endure extremely cold winters and seasonal variations in runoff. Additionally, through integrated outlets and spillways, the RCC structure facilitates multifunctional operation, enabling effective control of active storage, flood control, and ecological discharge.

4.1.5. ALIGNMENT WITH GERMAN ENGINEERING STANDARDS

The RCC dam concept complies with the German dam safety and storage classification standards outlined in DIN 19700. Thuringia has a history of managing its water resources responsibly, and the RCC design offers the flexibility and resilience needed to handle future water demand scenarios especially in light of anticipated population trends and climatic changes.

4.2. ECOLOGICAL DISCHARGE

The management of dam discharges is crucial when attempting to design and operate reservoirs, but it must also consider ecological impacts. These discharges affect not only the availability of water but also downstream ecosystem health. Without proper control, dam releases can cause reduced flow in the rivers; these releases may also affect aquatic life, such as grasslands, sediment transport, and species dependent on relatively stable water.

The continuous dumping of vast quantities of water for human consumption, including the drinking water supply, is one of the major environmental impacts. The consequence is frequently a prolonged reduction in natural streamflow, which can either increase the frequency and severity of low-flow periods or result in temporary flow disruptions. These circumstances may harm ecological integrity, resulting in losses of biodiversity and potential long-term benefits.

To maintain environmental protection while meeting human water needs, this project employs a seasonal approach to ecological discharge. During the summer season (April to September), the flow is maintained at 0.3 m³/s. Higher natural runoff from snowmelt and increased consumption are observed during this period. During the months of high demand, ecological processes like fish migration and riparian vegetation growth are maintained by ensuring minimal outflow.

An ecological flow rate of 0.15 m³/s is implemented during the winter season (October-March). This decrease is slight. A baseline flow that supports aquatic life during the colder, more dormant season is maintained by this conservative value, which accounts for lower demand and limited inflow due to freezing conditions.

These chosen values aim to maintain a sustainable balance, with enough water in the river to sustain ecosystems, and keeping the drinking water supply and other reservoir functions intact. These seasonal ecological discharges reflect hydrological patterns as well as environmental priorities, making the reservoir an ecologically responsible infrastructure not only of use but also of function.

Month	Ecological Discharge (m ³ /s)	Q(m ³ /d)	Days	Monthly Volume (m ³ /d)	Q (10 ⁶ m ³)
Jan	0.15	12960	31	401760,00	0,40
Feb	0.15	12960	28	362880,00	0,36
Mar	0.15	12960	31	401760,00	0,40
Apr	0.30	25920	30	777600,00	0,78
Mai	0.30	25920	31	803520,00	0,80
Jun	0.30	25920	30	777600,00	0,78
Jul	0.30	25920	31	803520,00	0,80
Aug	0.30	25920	31	803520,00	0,80
Sep	0.30	25920	30	777600,00	0,78
Oct	0.15	12960	31	401760,00	0,40
Nov	0.15	12960	30	388800,00	0,39
Dez	0.15	12960	31	401760,00	0,40

Table 2 : Monthly Ecological Discharge Volume Based on Seasonal Requirements

4.3. ACTIVE STORAGE

Active storage refers to the proportion of a reservoir's total storage volume that can be actively managed and utilized to meet water needs. The operational volume of the reservoir is situated between its minimum operating level and full supply level, which sets it apart from inactive or dead storage that may not be used due to technical or ecological limitations. Water availability and demand fluctuations in the short-term and seasonal periods are largely controlled by this volume. Active storage in the reservoir during excess inflow allows water to be captured and stored for use during drier months or peak demand periods, ensuring a steady and reliable water supply. Consistent discharge is essential for both human needs and ecological balance, particularly in areas prone to drought or with fluctuating seasons. [8]

Storage – English terms

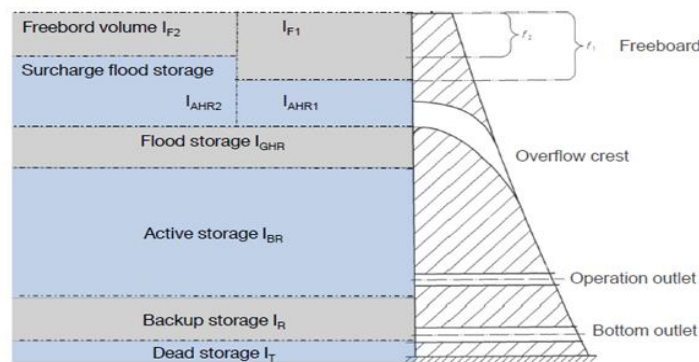


Figure 3: Reservoir Zonation Based on Storage Volumes

Water levels – English terms

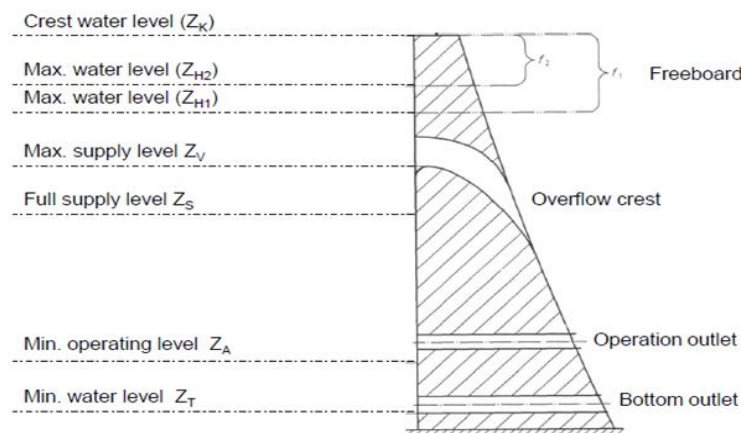


Figure 4: Reservoir Zonation Based on Water Levels

The Sequent Peak Algorithm (SPA) was employed in the current project for the calculation of active

storage. It is especially useful for determining the amount of storage necessary to compensate for changes in water flow over time. Flow data obtained from the Unterweißbach gauging station indicates that the discharge patterns of rivers were natural and not artificial. Due to the smaller size of the planned reservoir compared to its catchment area, an adjustment of 13.18% was applied to adjust the inflow data relative to that of the actual catchments.

$$\begin{aligned}\text{Reduction factor} &= 1 - \frac{A_2}{A_1} \\ &= 1 - \frac{72.6}{83.62} \\ &= \mathbf{13.18\%}\end{aligned}$$

A_1 = Catchment Area of Gauging Station

A_2 = Catchment area of dam

The outflow comprises the monthly water supply and Seasonal variations in the ecological discharge demand are reflected in both higher ecological and consumer demand, with a seasonal increase of 0.33 m³/s in summer and lower levels in winter (0.15 m³/sec) due to reduced needs and freezing. The population in the Thuringia region is experiencing fewer people (0.5%) than before, but this decrease was considered minor for the current planning horizon and, as a result, was not included in the adjustment to the demand. The approximate storage volume is a conservative estimate that guarantees durability under slight demographic changes.

From this method, we get the active storage volume as **8.42 * 10⁶ m³**. (Calculation and table are added in Appendix A)

4.4. INACTIVE STORAGE

A reservoir's inactive storage refers to the quantity of water that isn't utilized during normal operation, which is kept below the minimum operational level. The storage zone serves as a reserve strategic dam to enhance the dam's operational resilience in response to unexpected hydrological events such as extended droughts or significant drops. Inactive storage serves as a buffer, ensuring that critical water supply functions such as drinking water distribution and ecological flow maintenance can still be performed when

active storage is depleted or unavailable due to operational constraints. During times of low inflow or in case of emergencies, this reserve may be utilized to maintain water supplies and prevent supply disruptions, while also maintaining baseline flows downstream.

The project incorporated 7% of the active storage as an inactive storage volume. By considering the local hydrological characteristics and risk mitigation factors, this proportion was determined as a technically acceptable but conservative measure to avoid oversizing the reservoir. This allows for operational flexibility and ecological stability, which contributes to long-term water resource sustainability. [8]

4.5. DEAD STORAGE

The lowest part of a reservoir volume that is below the lowest outlet level and unavailable for routine operational use is referred to as dead storage. The main purpose of this storage section is to support sediment deposition during the dam's lifetime. Allocating dead storage is crucial to preventing sediment accumulation from lowering the effective capacity of the active storage zone because sediments carried by inflowing water naturally settle at the reservoir bed.

A dead storage height of 7 meters was set aside in the dam design for the current project. Given the complexity of sediment transport modeling, which usually necessitates comprehensive hydrodynamic and sediment load data for precise computation, this conservative estimate was selected. A fixed vertical buffer is designated as dead storage. This design maintains outlet functionality and long-term storage efficiency without requiring complex sediment simulation. [8]

4.6. FLOOD STORAGE

Flood storage is essential in reservoir design as it guarantees that the system can temporarily hold onto excess inflows during periods of heavy precipitation or flooding. Its main purpose is to reduce the risk of flooding downstream by controlling large inflow volumes and releasing water gradually after the peak has passed. Planning flood storage effectively helps safeguard ecosystems, infrastructure, and populated areas downstream of the dam.

This project uses a combination of unit hydrograph methods, design rainfall data (KOSTRA), and efficient precipitation estimation to determine flood storage. With the use of these hydrological tools, flood scenarios can be simulated to evaluate the reservoir's responsiveness and make sure the dam has enough capacity to withstand extreme events.

4.6.1. KOSTRA

To make calculations on designing rainfall hydraulic input, the precipitation information was obtained in the KOSTRA-DWD 2020R database using grid number 60045 that represents the position of the said project, including the coordinates 50.6006 °N and 11.1736 °E. KOSTRA is a system that gives standard rainfall intensity data of different durations and return periods across Germany. In the case of the given project, the amount of data was chosen as a 100-year return period ($T=100$) in view of an extreme precipitation event. It varies in intensity between 15.8 mm in 5 minutes, and 133.6 mm in 72 hours (4320 minutes). This time series volume of rainfall is important when considering the maximum flow values, the design of storage volumes during flooding and the assessment of the drainage system capacity. The findings are used in the continuation of rainfall-runoff simulation and quantification of storage volume during extreme climatic conditions. [9]

KOSTRA [T=100]	
DURATION [min]	PRECIPITATION [mm]
5	15.8
10	23.0
15	28.2
20	32.3
30	38.6
45	45.7
60	51.3
90	55.6
120	59.0
180	64.1
240	68.0
360	73.9
540	80.5
720	85.6
1080	93.3
1440	99.3
2880	120.4

4320	133.6
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Table 3 :Kostra Data

4.6.2. EFFECTIVE PRECIPITATION

The percentage of total rainfall that genuinely contributes to surface runoff, which eventually feeds into the reservoir or downstream systems, is known as effective precipitation. Losses from canopy interception, evaporation, infiltration, and other hydrological abstractions are not included. For reservoir design, it is essential to comprehend and precisely measure this component, particularly when calculating the true inflow volume from precipitation events.

The U.S. Soil Conservation Service (now NRCS) developed the SCS (Soil Conservation Service) method, which is used in this project to calculate effective precipitation from the total rainfall data acquired through KOSTRA. The SCS approach is especially successful because it uses a Curve Number (CN) to model rainfall-runoff relationships. This CN captures the combined impact of the catchment's antecedent moisture condition, land use, and soil type. The CN value, which represents the land cover and soil permeability characteristics of the region, is the primary catchment-specific parameter needed to apply the SCS method accurately and it ranges from 0 to 100. Using a standardized empirical formula, the method calculates the amount of precipitation that contributes to runoff once the CN and rainfall depth are known. Subsequent hydrological computations, including the unit hydrograph and flood storage design, use this effective precipitation value.

For the remaining calculations, it is necessary to have the following information about the catchment area.

- i. Soil type A, B, C, D
- ii. Antecedent Soil Wetness I, II, III
- iii. CN value

The effective precipitation N_{eff} can be calculated as a function of precipitation N and the CN- value with the help of the formula,

$$N_{eff} = \frac{\left[\frac{N}{25.4} - \frac{200}{CN} + 2 \right]^2}{\frac{N}{25.4} + \frac{900}{CN} - 8} \times 25.4$$

20

For the calculation, we must follow the steps below.

Determine Antecedent Soil Wetness

- i. Choose Soil Type
- ii. Determine CN-Value
- iii. Determine total N and effective precipitation N_D

An important factor in hydrological modeling is the Antecedent Soil Wetness Condition (AMC), which affects the soil's ability to infiltrate, and in turn, the amount of runoff produced by a precipitation event. Based on precipitation over the preceding five days, as well as seasonal and climatic factors, it describes the soil's initial moisture content prior to the start of rainfall. The Curve Number is greatly impacted by the choice of AMC: The CN value is lowered for AMC I to reflect its higher infiltration capacity. Since there is less infiltration and more potential for runoff, the CN is raised for AMC II. Therefore, accurate runoff volume estimation requires accurate antecedent soil moisture classification (N_{eff}). It guarantees that the catchment response to rainfall is neither overestimated in wet conditions nor underestimated in dry conditions. Usually, historical rainfall data, soil composition, vegetation cover, and temperature trends are used to inform AMC classification.

To ensure accurate flood routing and reservoir capacity estimation, the project's AMC determination was based on the rainfall history from the KOSTRA dataset, and the CN values were adjusted appropriately. Using Kostra data, the precipitation within the last 5 days is more than 50 mm. The soil is categorized as **Antecedent Soil Wetness II** due to the soil's ability to absorb additional rainfall is minimal, and runoff is likely to occur quickly. [9]

Soil wetness category	Precipitation within the last 5 days [mm]	
	vegetation period	other
I	< 30	< 15
II	30-50	15-30
III	> 50	> 30

Table 4 : Antecedent Soil Wetness Categories

Using QGIS software, which allowed for spatial referencing and hydrological response unit classification, the provided land use and soil data were analyzed to determine the catchment area's soil class. This geospatial assessment revealed that **clayey soil** was the predominant soil texture, along with a high groundwater table, which suggested a high potential for runoff and a low capacity for infiltration. The catchment area in this project is classified as belonging to the least permeable class, Hydrologic **Soil Group D**. Group D soils are very important for flood management and storage dimensioning because they usually produce the most surface runoff. Group D experiences significant effective precipitation (Neff) during rainfall events due to its low permeability and high antecedent moisture retention.

Soil Type	characteristics
A	soils with good infiltration ability, also after heavy moisturization, e. g. deep sand and gravel soils
B	soils with medium infiltration ability, deep to middle deep soils with modestly coarse texture, e. g. middle deep sand soils, loess or poorly loam-sand
C	soils with little infiltration ability, fine texture or water retaining layers, e. g. flat sand soils or sandy loam
D	soils with very little infiltration ability. Clayey soil, very flat soils above tight material; soils with a high ground water table

Table 5 : Soil Type

Using the table, the CN value (soil type) is determined depending on the land use (taken as an average) and soil type D, **CN 66** (Appendix B). Using the graph below, it is then adjusted depending on soil wetness category II.

Land use		CN-value (soil type)			
		A	B	C	D
Badland, waste land (without much plant cover)		77	86	91	94
Root crop, wine		70	80	87	90
Wine at terraces		64	73	79	82
Crop, animal feed		64	76	84	88
Pasture:	normal	49	69	79	84
	meager	68	79	86	89
Permanent grassland		30	58	71	78
Forest:	loose	45	66	77	83
	middle	36	60	73	79
	dense	25	55	70	77
Impermeable areas (e. g. sealed part of towns, roads etc.)		100	100	100	100

Table 6 : CN Values

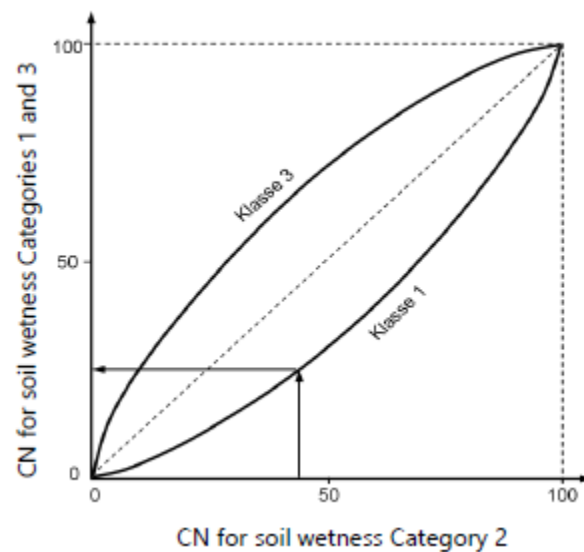


Figure 5 : CN Value Curve

Now the effective precipitation N_{eff} can be calculated from the equation,

$$N_{eff} = \frac{\left[\frac{N}{25.4} - \frac{200}{CN} + 2 \right]^2}{\frac{N}{25.4} + \frac{900}{CN} - 8} \times 25.4$$

The effective precipitation for each rainfall intensity is shown in the following table.

Duration (min)	Duration (hr)	N total (mm)	N _{effective} (mm)
5	0,08	15,8	0,89
10	0,17	23	0,08
15	0,25	28,2	0,03
20	0,33	32,3	0,27
30	0,50	38,6	1,08
45	0,75	45,7	2,54
60	1,00	51,3	4,05
90	1,50	55,6	5,40
120	2,00	59	6,59
180	3,00	64,1	8,52
240	4,00	68	10,13
360	6,00	73,9	12,76
540	9,00	80,5	15,94
720	12,00	85,6	18,56
1080	18,00	93,3	22,76
1440	24,00	99,3	26,22
2880	48,00	120,4	39,45
4320	72,00	133,6	48,44

Table 7: Effective Rainfall

4.6.3. UNIT HYDROGRAPH

A basic hydrological tool, the Unit Hydrograph (UH) shows how a catchment reacts over time to a unit pulse of rainfall, usually 1 mm/h dispersed evenly throughout the catchment area. It is particularly helpful for understanding the temporal behavior of streamflow, predicting runoff, and estimating peak discharge. The Unit Hydrograph was utilized in our hydraulic project to model the response of our planned catchment area to a typical rainfall event. We were able to estimate the amount of water that would eventually enter our reservoir and evaluate the timing of peak inflows, which is an important consideration for reservoir operation, flood control, and dam design.

The **Wackerman Method** was appropriate because less than 5% of our catchment area is sealed or impervious. We can represent both quick and slow runoff components using this method, which models the catchment as a system of two linear parallel cascades of reservoirs (representing different flow paths and delays). [9]

From our catchment analysis, we computed:

- i. Average River slope, $I = \frac{\Delta h}{L} = 0.02984$
- ii. Hydrographic Factor, $\frac{L}{\sqrt{I}} = 101.88 \text{ km}$
- iii. River density, $d = \frac{\Sigma L}{AE} = 1.2856 \text{ km/km}^2$
- iv. $K_1 = 4.48 - 2.25d = 1.5874 \text{ 1/h}$
- v. $K_2 = 2.50 + 0.0168 \cdot \frac{L}{\sqrt{I}} = 4.2117 \text{ 1/h}$

While it has a factor (dividing factor) β in the equation:

$$\beta = 0.323 \cdot e^{-0.00765 \cdot \frac{L}{\sqrt{I}}} = 0.1483$$

Finally, by applying the formula, we calculate the values of unit hydrograph, and we got it as shown in the figure below, (Details in Appendix B)

$$\begin{aligned} u(t) &= \beta \cdot u_1(t) + (1 - \beta) \cdot u_2(t) \\ &= t \left(\frac{\beta}{k_1^2} \cdot e^{t/k_1} + \frac{1-\beta}{k_2^2} \cdot e^{t/k_2} \right) \end{aligned}$$

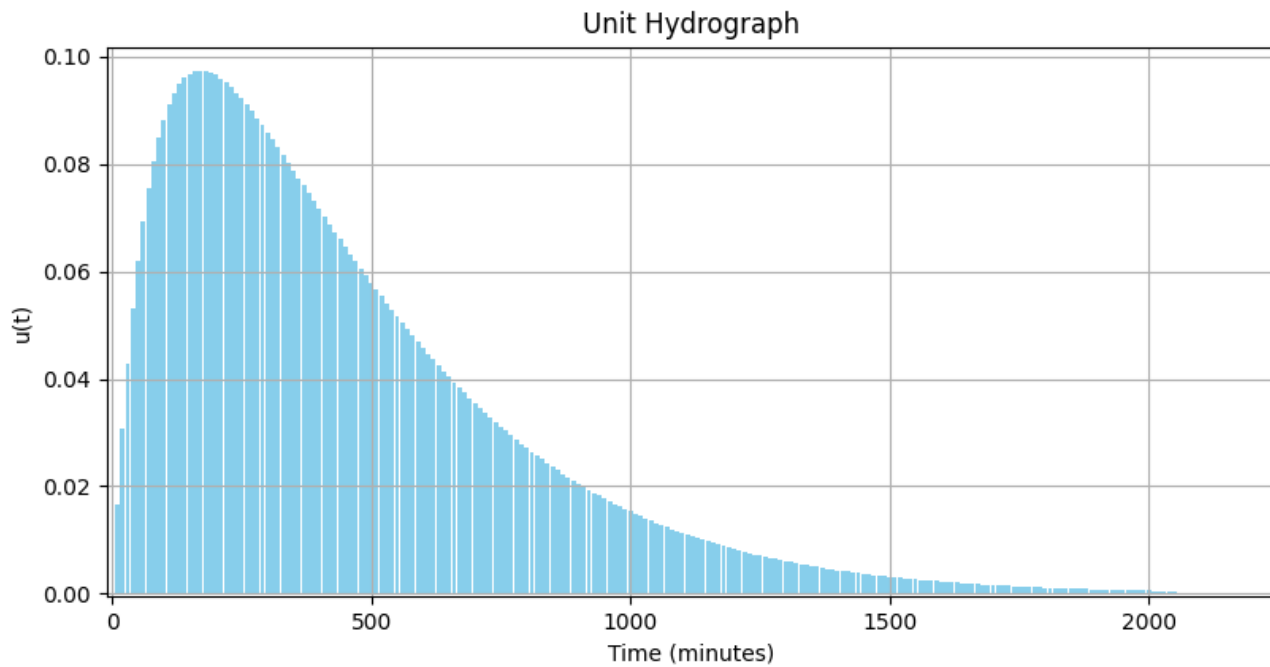


Figure 6 : Unit Hydrograph

4.6.4 REGULAR OUTFLOW

Regular outflow, which happens regardless of recent rainfall events, is the baseline discharge from the dam into the downstream river in dam hydrology and hydraulic engineering. This steady flow is essential for preserving the ecological balance downstream, sustaining aquatic habitats, and providing reliable water for industrial and municipal requirements.

The well-known Gauckler-Manning-Strickler formula, a hydraulic equation used to estimate open channel flow under steady-state conditions, was used in our project to calculate the regular outflow into the Lichte River. The formula is:

$$Q = k_{st} t * A * I^{\frac{1}{2}} * R_{hyd}^{2/3}$$

Where:

Q: regular flow discharge

k_{st} : Strickler coefficient

A: cross-sectional flow area

I: slope of the riverbed

R_{hyd} : hydraulic radius (ratio of area to wetted perimeter)

Let us assume that the slope of the riverbed is 0.9% and the width of the bridge is 3 m. So, the total area can be calculated as 3 m². Also, the hydraulic radius is equal to the height of water as it is considered a

rectangular one i.e. 1 m. Then the Stricker's value can be estimated at $k_{st} = 60 \text{ m}^{1/3}/\text{s}$. Considering all the above values the regular flow can be determined as, **17.08 m³/s**.

4.6.5. BASE FLOW

It is the constant movement of water from subterranean sources, such as groundwater passing through rock layers, soil, or crevices beneath the reservoir. Particularly during dry spells with little to no rainfall, this base flow is crucial. It keeps the reservoir's water level at a minimum so that, even in situations where surface water levels are low, supplies for things like drinking water, ecosystem requirements, and irrigation can continue.

The estimation of the amount of base flow that would enter our planned reservoir is conducted using a practical method. The steps include,

Several gauging stations in the catchment area were examined. River and stream flow levels are recorded at these monitoring locations.

The NNQ values, which represent the lowest typical flow and are frequently used as a base flow indicator, were gathered from each station.

We reduced the NNQ values to correspond with the size of our catchment because the CA for our dam is smaller than the areas those stations monitor.

We used the highest adjusted NNQ value as the base flow input to our reservoir just to be safe (it is **0.0976 m³/s** here).

Gauging Station	River	Area (km ²)	NNQ (m ³ /s)	Qbasis (m ³ /s)
Rudolstadt	Saale	2676	3.2	0.0862
Steinach	Steinach	37.8	0.02	0.0406
Blankenstein-Rosenthal	Saale	1013	0.308	0.0216
Camburg-Stöben	Saale	3973	5.2	0.0976
Mellingen	Ilm	627	0.14	0.0171
Unterebreizbach-Rasa	Ulster	396	0.16	0.0326
Arnstadt	Gera	174.5	0.2	0.0864
Schwarzburg	Schwara	340.6	0.243	0.0507

Table 8 : Gauging stations and catchment area

4.6.6. TIME DISTRIBUTION

The volume of rainfall events that are experienced differs in quantity as well as their distribution in time. This time-based distribution is useful in flood control planning since it helps know the rate at which water flows into a catchment, and the ability of a dam or reservoir to handle it safely. The analysis examines the relevance of the various rainfall durations and rainfall patterns on the runoff volume and the possible total volume of the flood. The objective of the organization of a particular scenario is to determine which of these scenarios is the most targeted in the development of flood storage safety and efficiency.

This part will review the impact of varying rainfall length and patterns of rainfall distribution on the volume of floods in the catchment. The catchment size taken in the analysis is 72.6 km², the base flow is 0.0976 m³/s, and the outflow capacity is 17.08 m³/s. The data on rainfall was obtained using the KOSTRA-DWD-2020 dataset with a 100-year return period, 6, 12, 24, 48, and 72 hours. The effective rainfall was found by deducting preliminary and constant losses off the total precipitation. The effective rainfall was further divided into a total of six equal time slots, with the patterns of four distributions: Uniform, Advanced, Intermediate, and Delayed. The patterns assume the possibility that rainfall intensity could vary with a storm event.

The quantities of runoffs were determined through the help of the unit hydrograph with each time slot, and they were compared to the capacity of discharge of the system. Each time the calculated flow was more than 17.08 m³/s, it was measured as the flood volume. This process was carried out at each of the durations and patterns to determine the most theoretically prominent situation. The maximum flood quantity was at the 48-hour Delayed pattern, where 2482376.36 m of superfluous water could occur. This value is the design value of the flood storage, and it is to be applied in the calculation of the capacity needed by the retention basin.

The detailed calculation of the time distribution can be found in Appendix C

DURATION (hr.)	UNIFORM	ADVANCED	INTERMEDIATE	DELAYED
6	1,618,914.06	1,649,371.34	1,672,079.04	1,644,303.99
9	1,862,955.10	1,915,118.67	1,947,457.70	1,887,622.85
12	2,014,529.56	2,092,440.58	2,139,617.00	2,054,809.38
18	2,150,276.20	2,257,550.82	2,345,027.32	2,232,844.79

24	2,157,516.43	2,273,211.74	2,418,569.01	2,309,318.16
48	1,845,462.98	1,986,611.57	2,337,567.37	2,482,376.36
72	942,923.42	1,809,278.80	2,083,834.55	2,266,558.05

Table 9 : Time Distribution of the Study Area

4.7 SURCHARGE FLOOD STORAGE

The temporary extra volume that a reservoir can temporarily hold above its typical full supply level (FSL) in order to safely handle extreme flood events is known as surcharge flood storage. When the dam's active and buffer storage capacities are exceeded by unusual rainfall or flood conditions, this storage becomes vital. Normally, it is not used for drinking water, irrigation, or power generation. The maximum water level (MWL) that the dam can structurally support without overtopping is reached by this extra storage zone, which is situated above the maximum conservation level. To absorb peak flood discharges over brief periods of time, it serves as a safety margin, lowering flood peaks downstream, avoiding structural overloading as well as allowing time for outlet gates or spillways for controlled releases. Put more simply, consider surcharge flood storage to be a reservoir's emergency overflow room, a vital buffer zone that safeguards the dam and communities downstream in the event of exceptionally high floods.

4.7.1 GUMBEL METHOD

Distribution functions are essential tools in hydrology, especially for spillway design of large dams, where extreme flood probabilities are critical. Return periods for spillway design of big dams: $T = 1000 / 10000$ y to ensure structural safety. Flood data series are typically data series max. ≈ 100 y.

Gumbel Extreme value Type 1 distributions were employed in this project design to estimate the surcharge flood of BHQ1 and BHQ2 for the return period of 1000 years and 10000 years respectively. The data used is from 1969 to 2019. The yearly peak discharge recorded at the gauging station Unterweißbach was utilized for the purposes of calculation.

We have employed the Gumbel method to calculate the usual flood storage.

$$HQ_T = MHQ + \Delta HQ_T$$

$$HQ_T = MHQ + \sigma_{HQ} k(n, T)$$

The parameters of the equation, such as mean discharge, and standard deviation, are calculated from the provided table by using the following formula.

Year	Discharge [m ³ /s]
1969	22.3
1970	14.3
1971	17.5
1972	8.4
1973	26.5
1974	15.2
1975	14.2
1976	11.8
1977-2019	19.4-17.11

Table 10 : Year and corresponding discharge

Step 1: MHQ

$$MHQ = \frac{1120.8}{51} = 21.98 \text{ m}^3/\text{s}$$

Step 2: $\sigma_{HQ} = 12.56 \text{ m}^3/\text{s}$

Calculating the data, from the table provided we get the average discharge of 21.98 m³/s and a standard deviation of 12.56 m³/s.

Using the formulas below, we can determine the coefficient KG based on the number of years (n= 51) and the year of the return period, depending on the discharge.

$$K_G(n, T) = \frac{y_T - \bar{y}_n}{s_n}$$

$$y_T = -\ln\left(-\ln\left(1 - \frac{1}{T}\right)\right)$$

(1) Arithmetic mean $m = MHQ = \frac{1}{n} \cdot \sum_{i=1}^n HQ_i$

(2) Standard deviation $s = \sigma_{HQ} = \sqrt{\frac{1}{n-1} \cdot \sum_{i=1}^n (HQ_i - MHQ)^2}$

(3) skewness $c_s = \frac{n \cdot \sum_{i=1}^n (HQ_i - MHQ)^3}{(n-1) \cdot (n-2) \cdot \sigma_{HQ}^3}$

The parameters Y_n and S_n can be determined from a table using the following formula

$$\bar{y}_n = \frac{1}{n} \cdot \sum_{i=1}^n y_i = \frac{1}{n} \cdot \sum_{i=1}^n -\ln\left(-\ln\left(1 - \frac{1}{i}\right)\right)$$

$$s_n = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_n)^2}$$

Calculations

$$y_{51} = \frac{1}{51} \cdot i$$

$$s_n = S_{51} = \sqrt{\frac{1}{51} \cdot \sum (y_i - 0.5426)^2} = \mathbf{1.1124}$$

BHQ₁ T = 1000 years

$$yT = -\ln\left(-\ln\left(1 - \frac{1}{1000}\right)\right) = -\ln(0.00100005) = 6.9072$$

$$K = \left(6.908 - \frac{0.5426}{1.1124}\right) = 5.722$$

$$\mathbf{BHQ_{1000} = 21.98 + (12.56 \cdot 7.792) = 21.98 + 71.89 = 93.87 \text{ m}^3/\text{s}}$$

BHQ₂ T = 10,000 years

$$yT = -\ln\left(-\ln\left(1 - \frac{1}{1000}\right)\right) = -\ln(0.000100005) = 9.210$$

$$\mathbf{K} = \left(9.210 - \frac{0.5426}{1.124}\right) = 7.792$$

$$\mathbf{BHQ_{10000} = 21.98 + (12.56 \cdot 7.792) = 21.98 + 97.86 = 119.84 \text{ m}^3/\text{s}}$$

BHQ₃ T = 100 years

$$yT = yT = -\ln\left(-\ln\left(1 - \frac{1}{1000}\right)\right) = -\ln(0.001005) = 4.60$$

$$K = \left(4.60 - \frac{.5426}{1.1124}\right) = 3.674$$

$$\mathbf{BHQ100} = 21.98 + (12.56 * 3.674) = 21.98 + 45.833 = \mathbf{67.81m^3/s}$$

Calculating from the equations above, the surcharge floods of **BHQ1** and **BHQ2** are determined as **93.87m³/s** and **119.84m³/s**, respectively. **BHQ3** is the same as HQ100, which is **67.81m³/s**.

4.8 WATER LEVEL- VOLUME RELATIONS

The reservoir's storage curve is given by the relationship between water level and storage capacity over time. This curve helps in understanding the dam's performance and operational characteristics, as it gives information about water availability, flood control capacity, and potential for other uses such as irrigation, hydropower generation, and supplying drinking water.

The storage curve was developed using QGIS. The "Raster Surface Volume" tool, found under Raster Analysis, was used to calculate the volume and surface area of the reservoir at various water levels. The Digital Elevation Model (DEM) of the reservoir served as the input layer. The option "Count only below base level" was selected to ensure that calculations considered only the volume beneath each specified water level. Using this method, the volume and surface area were computed incrementally for a range of base levels, resulting in the data needed to construct the storage curve.

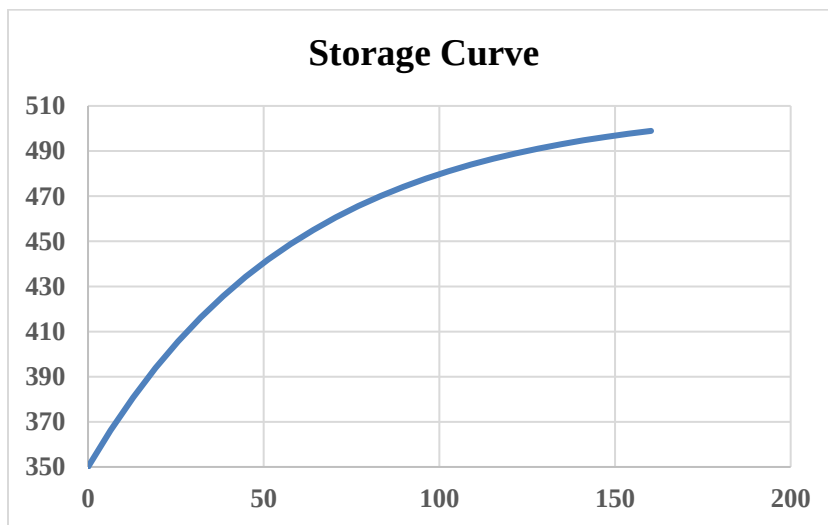


Figure 7 : Storage Curve

4.8.1 STORAGE LEVELS AND VOLUMES

In the previous section, the storage curve generated earlier was used to determine the storage volumes at various water levels by interpolating the curve values. The dead storage was set at a height of 7 meters, this volume is intended to accommodate sediment accumulation over time. The inactive storage volume was determined as 7% of the active storage volume, corresponding to a height of 9.6 meters, as derived from the storage curve. The active storage and normal storage volumes were also calculated based on the same method.

Storage	Volume (m³)	Equivalent height (m)
Dead Storage	457701	7.0
Active Storage	8424960	38.2
Inactive Storage	589747.2	9.6
Flood Storage	2409906	20.19

Table 11 : Storage Volume

To estimate the surcharge flood storage, the equivalent dam height was determined using the Poleni weir flow equation, assuming a spillway width of 10 meters.

$$Q = \frac{2}{3} \mu \sqrt{2g} B h^{3/2}$$

Overflow depths for BHQ_1 is 93.87 m³/s and BHQ_2 is 119.84 m³/s were determined. The corresponding equivalent storage volumes were calculated using these overflow depths.

Design flood discharge	m ³ /s	Equivalent height (m)
BHQ_1	457701	2.61
BHQ_2	8424960	3.08

Table 12 : Design flood Discharge

4.9 FREE BOARD

The free board is a critical design feature of the RCC dam, ensuring that the crest remains safely above the reservoir's maximum water level during extreme loading conditions such as flood events, strong wind influence, ice built up and operational uncertainties.

To determine the required freeboard, a compact model is applied in accordance with DIN 19700 [2] and supported by the methodologies outlined in [10]. This method considers the combined effects of:

1. Wind-induced wave run up of the reservoir surface
2. Wind setup height (quasi-static rise in water surface due to wind pressure)
3. Ice jam formation
4. A safety surcharge

Two design scenarios are evaluated based on the return period of the wind event during the flood:

Case I: Wind Return period $T_{Wind} = 25$ years

$$f_1 = h_{Au} + h_{\omega i} + h_{Ei}$$

Case II: Wind Return period $T_{Wind} < 25$ years

$$f_2 = h_{Au} + h_{\omega i} + h_{Ei} + h_{Si}$$

The runoff waves h_{Au} was determined to be 1.2 m using the Krylov II approach. Water level increase $h_{\omega i}$ (windstau) was taken to be 0.05m from the table below for a fetch of 2000m. [11] The ice jam h_{Ei} (Eisstau) and safety margins were assumed to have a value of 0.5 m and 0.4 m each and the free boards were obtained as 1.75 m and 2.15 m, respectively.

Given the strategic importance and safety classification of the structure, and considering that precise local wind recurrence data may not be available with high certainty, the conservative 2 is adopted for design

Load Case	Freeboard	Height (m)
Case 1	f_1	1.75
Case 2	f_2	2.15

Table 13 : Freeboard Height

The calculation is given in Appendix E

4.10 DAM GEOMETRY

The maximum water levels, Z_{H1} and Z_{H2} , were determined by adding the respective freeboard of 2.15 to the water levels associated with BHQ_1 and BHQ_2 . For the final design, the crest water level is Z_{K2} . The storage level of the dam is shown in the following table.

Elevation (masl)	Water level (m)		Storage volume (10 ⁶ m ³)		Dimension (m)
428.15	Crest level (Z _{K2}) (BHQ2+F)	82.13	Freeboard		2.15
426	Maximum water level (Z _{H2})	78.6	Surcharge flood storage I _{AGHR} (BHQ ₂)	0.26	3.08
424.20	Maximum water level (Z _{H2})	71.6	Surcharge flood storage I _{AGHR} (BHQ ₁)	0.24	2.61
402.01	Maximum supply level (Z _V)	69.4	Usual flood storage I _{GHR}	0.240	20.19
400	Full supply level (Z _S)	57.6	Active storage I _{BR}	8.42	38.2
379.6	Minimum operation level	14.6	Inactive storage I _R	0.589	9.6

	(Z _A)				
360	Minimum water level (Z _T)	7.00	Dead storage I _T	0.4577	7.0
350	Dam bottom				

Table 14 : Dam Geometry

The height of the dam is determined to be 78.15. The crest width (T) is also calculated as 7.09 m, and 7 m is adopted. The following equation is used to measure crest width

$$T = 1 \cdot 65(H + 1.5)^{0.33}$$

To classify the dam's structural safety and determine whether it should be classified as a low or high dam, the limiting height is obtained as 88.72 m and was calculated using a standard empirical formula.

$$H_{\text{lim}} = \frac{f}{\gamma_w(S_c + 1)}$$

where $f = 3000 \text{ kN/m}^3$ is the allowable compressive stress of concrete, γ_w is the unit weight of water 9.81 kN/m^3 and S_c is 2.446. Since the actual dam height of 78.15 m is less than the limiting height, the dam is considered a low dam in accordance with DIN 19700 classification. [2]

.To ensure dam stability, especially against sliding and overturning forces. The required base width was also calculated using the following formula

$$B = H \left(\frac{S_c + 1}{2S_c} \right)$$

Using the previously calculated values and the actual height, the base width is 55.05 m. Combined with the adopted crest width of 7.0 m, the cross-section profile of the dam ensures both hydraulic and design requirements.

5.0 OPERATIONAL FACILITIES

Operational facilities in a dam are essential components that enable the safe, efficient, and controlled management of water storage and release. These include structures such as bottom outlets, operational outlets, spillways, control gates, and valves, which together regulate reservoir levels, ensure water supply for domestic and agricultural use, and provide emergency discharge capacity during flood events. Properly designed and maintained operational facilities support the day-to-day functionality of the dam, ensure structural safety, and help maintain downstream flow continuity for ecological balance and public needs

5.1 RESERVOIR ROUTING

Reservoir routing describes the methods of computing an outflow hydrograph given an inflow hydrograph and reservoir characteristics. According to the currently valid standards (DIN 19700) spillways may be designed considering the retention effect ($Q_{out} < Q_{in}$). Reservoir routing is used to determine the relationship between the storage capacity of the reservoir, the inflow into the reservoir, and the outflow of the reservoir during the flood. The retention effect of the dam is demonstrated by the relationship between the reservoir's storage capacity, inflow, and outflow (e.g., by spillway and/or bottom outlet).

The Puls method, also known as the storage-indication method, is a widely used hydrologic technique for flood routing through reservoirs or detention basins. It is based on the principle of mass conservation and relates inflow, outflow, and storage over time. The method assumes a known relationship between storage and outflow and uses a step-by-step approach to calculate how an inflow hydrograph is transformed by the reservoir. By balancing the change in storage with the difference between inflow and outflow over each time interval, the Puls method allows engineers to estimate the attenuated and delayed outflow hydrograph. It is especially useful in dam and reservoir design, where understanding peak reduction and flood wave timing is critical for safety and efficiency.

Puls Method Mathematical Equations: The Puls method is based on some mathematical equations that show the relationship between the inflow hydrograph and the outflow, thereby considering the retention effect of the reservoir. [12]

$$\frac{dS(t)}{dt} = Q_i(t) - Q_o(t)$$

↓ integrate approximatively

$$\frac{S(t+\Delta t) - S(t)}{\Delta t} = \frac{Q_i(t) + Q_i(t+\Delta t)}{2} - \frac{Q_o(t) + Q_o(t+\Delta t)}{2}$$

↓ sort by **unknown** and **known** values

$\frac{2S(t+\Delta t)}{\Delta t} + Q_o(t+\Delta t)$

$= \frac{2S(t)}{\Delta t} - Q_o(t) + Q_i(t) + Q_i(t+\Delta t)$

$Q_i(t) + Q_i(t+\Delta t)$

e.g. forecast, given hydrograph, ..

$S = f(W)$
 $Q_o = f(W)$

5.2 BOTTOM OUTLET

The bottom outlet of a dam serves as a critical structural component for managing reservoir operations, particularly in regulating water levels and ensuring controlled outflows. It allows for the release of water during normal and emergency conditions, thereby preventing overtopping, reducing flood risks, and maintaining downstream ecological flows. In this project, the bottom outlet is specifically designed to discharge regular ecological flow while supporting reservoir maintenance and operational flexibility. The outlet features a rake system and a gated control mechanism, enabling flow regulation and inspection access when necessary.

The rake, positioned at the intake, plays an essential role in debris management by preventing the entry of obstructive materials such as branches, leaves, and larger floating objects. This protective measure safeguards the intake structure from clogging, which can compromise discharge efficiency, reduce flow capacity, and elevate the risk of local flooding or structural damage. Additionally, bottom outlets support sediment management by enabling periodic pressure flushing, which helps to minimize sediment buildup near the dam and downstream riverbed. This is vital for maintaining the hydraulic performance of the reservoir and ensuring a balanced downstream ecosystem.

5.2.1 DISCHARGE CALCULATIONS

$$Q = A_A \cdot \sqrt{2g} \cdot \sqrt{\frac{h_E - \left(\frac{p_A}{\rho g}\right)}{1 + \sum \text{energy losses}}}$$

Q = Discharge (m³/s)

A_A = Cross-sectional area of the outlet (m²)

g = Acceleration due to gravity (9.81 m/s²)

h_E = Available energy head (m)

p_A = Atmospheric pressure

P = Water density (kg/m³)

The area will be designed considering some losses such as rake, inlet losses, and frictional losses.

$$\sum \text{losses} = K_{\text{rake}} + K_{\text{inlet}} + K_{\text{friction}}$$

Head Loss Through Bar Racks

$$K_{\text{rake}} = \beta^2 * \left(\frac{t}{s}\right)^2 * (\sec \alpha)^2$$

Where:

h_L : Head loss (m)

β: Empirical coefficient (depends on bar shape, angle, and blockage)

t: Bar thickness (m)

s: Bar spacing (m)

α: Installation angle (degrees, typically from vertical or flow direction)

v: Approach flow velocity (m/s)

g: Gravitational acceleration (9.81 m/s²)

Assuming the following values for the calculation:

Variables	Values
Bar thickness (d)	25 mm
Spacing (s)	100 mm

Installation angle (α)	90°
β	1.83 (rectangular bars at 90°)
V (assumed)	2 m/s
g	9.81

Table 15 : Parameters

Step-by-Step Calculation

Krake:

$$= \beta^2 * \left(\frac{t}{s}\right)^2 * \sec^2(\alpha)$$

Since $\alpha=90^\circ$

$$= \sec(90^\circ) = \frac{1}{\cos(90^\circ)} = \frac{1}{0}$$

But in real engineering practice, a 90° bar means bars are perpendicular to flow, so $\sec(\alpha)$ is typically treated as 1.

$$= (1.83)^2 \cdot (0.25)^2 \cdot 1 = (3.35 * 0.0625) = 0.209$$

K inlet: Inlet loss:

Typically for a sharp-edged entrance, **K=0.5**

K friction: Friction loss:

We are assuming **K=0.5** for a short pipe section

$$\sum \text{losses} = 0.209 + 0.5 + 0.5 = 1.209$$

Further Calculations

$$Q = A_A \cdot \sqrt{2g} \cdot \sqrt{\frac{h_E - \left(\frac{p_A}{\rho g}\right)}{1 + \sum \text{energy losses}}}$$

Substituting the values to find Area and Diameter

$$Q = 17.08 \text{ m}^3/\text{s}$$

$$h_E = 62.4 \text{ m}$$

$$g = 9.81 \text{ m/s}^2$$

$$\sum \text{losses} = 1.209$$

Pressure $p_A = 0$ (atmospheric)

Rearranging the equation

$$A_A = \frac{Q}{\sqrt{2g} \cdot \sqrt{1 + \sum K}}$$

Substituting the values

$$= \frac{17.08}{\sqrt{2} \cdot 9.81 \cdot \sqrt{(62.4/(1 + 1.209))}}$$

$$= \frac{17.08}{(4.429 \cdot 5.315)}$$

$$= \mathbf{0.726 \text{ m}^2}$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$= \sqrt{\frac{(4 \cdot 0.726)}{3.14}}$$

$$= \mathbf{0.961 \text{ m}}$$

Calculated values and variables are structured in the following table

Variables Calculated	Value
Q (regular outflow)	17.08 m ³ /s
h_E	(69.4 – 7 = 62.4) m
g	9.81 m/s ²
Area calculated	0.726 m ²
D calculated	0.961 m
D adopted	1.1m

Table 16 : Variables Calculated

5.3. OPERATIONAL OUTLET

The operational outlet of a dam is a vital component of its water management system, designed to safely and efficiently release water from the reservoir to meet specific, planned objectives. In this case, its primary function is to supply drinking water, but operational outlets often serve multiple roles, including flood control, irrigation, environmental flow maintenance, and even hydropower generation. To fulfill these roles, the outlet must be engineered to handle the maximum expected discharge, with sufficient capacity to respond to both routine demands and emergency scenarios. It should be equipped with control gates, valves, and monitoring instruments to precisely regulate flow rates under varying reservoir levels and operational conditions. The strategic placement of operational outlets is critical typically located at elevations that allow water withdrawal across a range of storage levels, depending on their intended use. For example, drinking water outlets may be placed at intermediate levels to avoid sediment-laden bottom layers and ensure water quality, while outlets used for flood control may be situated lower in the reservoir.

Routine maintenance of the operational outlet is essential to ensure uninterrupted functionality, as blockages, mechanical failures, or sediment accumulation can severely limit the dam's ability to release water effectively. Proper operation and upkeep of these outlets not only support reliable water delivery for downstream uses but also contribute significantly to the overall safety and efficiency of the dam infrastructure.

In this project, the operational outlet system was specifically designed to supply drinking water to nearby communities. Two separate outlets were incorporated into the design: one to accommodate the monthly water demand during regular operations, and another reserved for emergency situations. Both outlets are positioned at the same elevation of 379.6(masl), corresponding to the minimum operating level of the reservoir.

During the design process, several critical hydraulic and structural factors were considered to ensure the outlets function safely and efficiently. Key considerations included the potential for cavitation, pressurized flow behavior, and the need for flow regulation mechanisms. These elements are essential to maintaining long-term performance, minimizing the risk of damage, and ensuring reliable water delivery under varying operating conditions.

Step-by-Step Calculation

$$Q = A_A \cdot \sqrt{2g} \cdot \sqrt{\frac{h_E - \left(\frac{p_A}{\rho g}\right)}{1 + \sum \text{energy losses}}}$$

$$\sum \text{losses} = K_{\text{inlet}} + K_{\text{valve}} + K_{\text{bend}} + K_{\text{friction}} + K_{\text{exit}}$$

Assuming the same value as the bottom outlet and K_{valve} as 1 and K_{exit} loss as 1

$$\sum \text{losses} = 0.5 + 0.5 + 0.5 + 1 + 1 = 3.5 \text{ m}$$

$$Q = 1.83 \text{ m}^3/\text{s}$$

$$h_E = 47.8 \text{ m}$$

$$g = 9.81 \text{ m/s}^2$$

$$\sum \text{losses} = 3.5$$

Outlet pressure $p_A = 0$ (atmospheric)

Rearranging the equation

$$A_A = \frac{Q}{\sqrt{2g} \cdot \sqrt{\frac{h_E}{1 + \sum K}}}$$

$$= \frac{1.83}{\sqrt{2} * 9.81 \cdot \sqrt{\left(\frac{47.8}{1 + 3.5}\right)}}$$

$$= \frac{1.83}{(4.429 * 3.259)}$$

$$= 0.127 \text{ m}^2$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$D = \sqrt{\frac{(4 \cdot 0.127)}{3.14}} = \mathbf{0.401 \text{ m}}$$

The design of the operational outlet follows a similar methodology to that of the bottom outlet, including the consideration of hydraulic losses, which were calculated to be 3.50 meters. The required discharge was determined based on the maximum monthly water demand, which occurs in July, with a total volume of $4.88 \times 10^6 \text{ m}^3$. This was converted into a continuous discharge rate of $1.83 \text{ m}^3/\text{s}$. For the head driving the flow (h_E), the total height of the active and inactive storage zones was used, giving an effective head of 47.8 meters. Based on these parameters and loss considerations, a pipe diameter of 0.401 meters was initially calculated. However, for design and operational flexibility, a standard diameter of 0.5 meters was ultimately adopted.

Calculated values and variables are structured in the following table

Variables	Value
Q	$1.83 \text{ m}^3/\text{s}$
h_E	$(38.2 + 9.6 = 47.8) \text{ m}$
g	9.81 m/s^2
Area calculated	0.127 m^2
D calculated	0.401 m
D adopted	0.5 m

Table 17 : Calculated Variables and Values for Operational Outlet

5.4. SPILLWAYS

The spillway is a critical safety feature of any dam, designed to provide a controlled path for the release of excess water from the reservoir. Its primary function is to prevent overtopping of the dam during periods of heavy rainfall, rapid snowmelt, or extreme flood events. Without a properly designed and functioning spillway, a dam can be subjected to uncontrolled water levels, which may lead to overtopping—a dangerous scenario that can result in the erosion of the dam's crest and downstream

face, ultimately compromising the structural integrity of the dam and potentially leading to catastrophic failure.

Spillways are typically engineered to accommodate the probable maximum flood (PMF) or other design flood criteria, ensuring that the dam can safely discharge large volumes of water without sustaining damage. By maintaining the reservoir level within safe operating limits, spillways reduce hydrostatic pressure on the dam body and foundation, thereby contributing to the overall stability and longevity of the structure.

Additionally, the controlled release of water through the spillway protects downstream communities, ecosystems, and infrastructure from sudden flooding. Modern spillway designs may include features such as gates, energy dissipation basins, and aeration systems, all aimed at optimizing flow regulation and minimizing erosion or cavitation.

In summary, a well-designed and maintained spillway is essential not only for the operational efficiency of the dam but also for ensuring public safety and environmental protection. The spillway system in this project is designed using the Pollini equation, assuming an ideal overflow profile with a discharge coefficient of $C = 0$. The structure is classified as an uncontrolled spillway; it operates without gates or mechanical regulation, allowing water to flow freely once it exceeds the crest level. The design follows the commonly adopted rule, which ensures that the spillway system can still function effectively even if one spillway becomes inoperative.

To meet the required discharge capacity and enhance operational reliability, two spillways will be constructed. Under normal conditions, the primary spillway will handle regular flood discharges. The secondary spillway is intended to serve as an emergency or backup structure, activated in extreme flood events or when maintenance is required on the primary spillway. This redundancy is crucial for maintaining safe reservoir levels during critical conditions and ensures continuous functionality without compromising dam safety.

Step-by-Step Calculation

Bottom Outline (Torricelli equation)

$$Q_{0, \text{bottom outlet}} = \varphi \cdot A \cdot \sqrt{2g} \cdot (W - H_{\text{bottom outlet}})^{1/2}$$

Spillway (Pollini equation)

$$Q_{0, \text{spillway}} = \frac{2}{3} \cdot \mu \cdot b \cdot \sqrt{2g} \cdot (W - H_{\text{spillway}})^{3/2}$$

$$Q \text{ (BHQ2)} = 119.84 \text{ m}^3/\text{s}$$

$$\mu = 0.75$$

$$g = 9.81 \text{ m/s}^2$$

$$b = 7$$

$$H = 3 \text{ m}$$

$$\text{i.e. } 119.84 = \frac{2}{3} \cdot 75 \cdot b \cdot (\sqrt{2} \cdot 9.81) \cdot 3^{\frac{3}{2}}$$

$$119.84 = 11.206b$$

$$\mathbf{b = 10.41 \text{ m}}$$

Recalculating Q new with adopted b value

$$Q_{\text{new}} = \frac{2}{3} \cdot 0.75 \cdot 12 \cdot (\sqrt{2} \cdot 9.81) \cdot 3^{\frac{3}{2}}$$

$$Q_{\text{new}} = \mathbf{138.08 \text{ m}^3/\text{s}}$$

Calculated values and variables are structured in the following table

Variables	Value
Q (BHQ₂ Puls)	119.84 m ³ /s
μ	0.75
g	9.81 m/s ²
h	3.0 m
B (calculated)	10.41 m
B (adopted)	12 m
Q_{new}	138.08 m³/s

Table 18 : Variables of Spillway Design

5.5 ENERGY DISSIPATION STRUCTURES

Energy dissipation structures are essential components of dam design, serving the critical function of reducing the kinetic energy of water discharged from the dam to prevent erosion, structural damage, and environmental degradation in downstream areas. When water exits the spillway at high velocity, the resulting hydraulic forces can cause severe scouring at the dam toe, destabilize riverbanks, and

compromise both the dam structure and downstream ecosystems.

To mitigate these risks, a variety of energy dissipation structures are employed, including stilling basins, baffle blocks, chute blocks, flip buckets, and stepped spillways. These are typically constructed at the toe of the spillway or within a re-regulation basin, where they serve to slow the flow and safely disperse energy over a wider area.

5. 5.1 STEPPED SPILLWAY DESIGN IN THIS PROJECT

In this project, energy dissipation is achieved through the incorporation of a stepped spillway constructed directly over the (RCC) dam. The use of stepped spillways is a widely adopted and effective method for energy dissipation in both gravity and embankment dams. This approach not only regulates the flow of water during flood events but also reduces the need for additional downstream energy dissipation structures. A stepped spillway consists of a series of horizontal steps or drops along the chute, which disrupt the flow and induce turbulence and air entrainment. These steps significantly increase flow resistance and promote energy loss through turbulence and surface drag. The entrained air also cushions the flow and reduces cavitation risk, further protecting the dam's structure. By dissipating a significant portion of the flow energy along the chute, the stepped spillway design minimizes the necessity for larger and more complex stilling basins or flip buckets at the downstream end. Additionally, this design supports safer flood passage, improves hydraulic performance, and contributes to the overall stability and longevity of the dam infrastructure.

Step-by-Step Calculation

1)Critical Depth h_c :

$$h_c = \left(\frac{q^2}{g} \right)^{1/3}$$

h_c = critical depth (m)

q = unit discharge = Q/B (m^2/s)

Q = total discharge (m^3/s)

B = channel width (m)

$g = 9.81 \text{ m/s}^2$

$$q = \frac{138.08}{12} = 11.50$$

$$hc = \left(\frac{(11.50)^2}{9.81} \right)^{\frac{1}{3}} = 2.379$$

2) Step height S

Recommended step height range is often based on critical depth h_c

Using DIN 19700 and DVWK-M 238

$$1.43m \leq S \leq 3.57m (\text{based on } h_c = 2.379 m)$$

Assuming S as **3 m** from these standards.

3) Uniform clear water depth $H_{w,u}$

For stepped spillways, the uniform clear water depth under skimming flow is typically less than the critical depth (Chanson 1994, DVWK- (Chanson 1994, DVWK-M 238)

$$H_{w,u} / h_c \approx 0.25 - 0.3$$

$$H_{w,u} / h_c = 0.25$$

$$H_{w,u} = 0.595m$$

4) Aeration length L_i

$$L_i = \beta \cdot h_c \cdot \left(\frac{S}{h_c} \right)^{0.5}$$

empirical correlation (Chamani & Rajaratnam, 1999)

β = empirical constant 8.5–9.0 for RCC stepped spillways

h_c = critical depth

S = step height

$$(S/h_c)^{0.5} = \left(\frac{3}{2.38} \right)^{0.5} = 1.123$$

$$L_i = 8.5 \cdot 2.38 \cdot 1.123 = 22.72m$$

5) Water depth at 90% of air entrainment $H_{90,u}$

According to Chanson (1997, 2002) and DVWK-M 238

$$H_{90,u} \approx 2.33 \cdot H_{w,u}$$

$$H_{90,u} = 2.33 \cdot 0.595 = \mathbf{1.386m}$$

6) Leading wall height h_d

$$h_d = H_{90,u} + \text{freeboard}$$

freeboard = safety margin (0.2 to 0.3 m, or 10–20% of $H_{90,u}$)

$$h_d = 1.386 + 0.30 = \mathbf{1.69m}$$

7) Flow depth at aeration start

$h_i \approx 1.88 \cdot H_{w,u}$ based on Chanson's experimental findings (1994–2002)

$$h_i = 1.88 \cdot 0.595 = \mathbf{1.12m}$$

8) Residual energy

$$\frac{H_{res}}{H} = 1 - \varepsilon$$

$$\varepsilon = 0.55 + 0.15 \sqrt{\frac{S}{h_c}}$$

From Chanson (2002) and DVWK-M

$$= \left(\frac{3}{2.38} \right)^{0.5} \times 0.15 + 0.55 = 0.72$$

$$H_{res} = (1 - 0.72) \cdot 100 = 28\%$$

This design emphasizes the importance of stepped spillways for effective energy dissipation and, as such, incorporates a cascade-type chute spillway into the dam structure. The key design characteristics of the chute spillway are summarized in the table below:

Parameter	Symbol	Value
Critical depth	h_c	2.38 m

Step height	S	3.00 m
Uniform clear water depth	$H_{w,u}$	0.595 m
Aeration Length	L_i	22.72 m
Water depth at 90% of air entrainment	$H_{90,u}$	1.386 m
Leading wall height	h_d	1.69 m
Flow depth at aeration starts	h_i	1.12 m
Residual energy	H_{res}	28%

Table 19 :Parameters for Stepped Spillway

6.0 STABILITY ANALYSIS

To make sure an RCC dam can sustain all acting forces, such as hydrostatic pressure, uplift, self-weight, seismic activity, and foundation interactions, stability analysis is crucial. RCC dams depend on their mass and contact with the foundation for stability because of their inflexible, monolithic design. Using Limit Equilibrium techniques, the analysis looks for sliding, overturning, and bearing capacity failure; safety factors are usually greater than 1.5 for dependability. Although there is little deformation because of the rigid construction, special attention is paid to the integrity of the foundation and construction joints to ensure long-term performance and safety.

6.1 FORCES ACTING ON THE DAM

Gravity dams are designed to use their own weight and structural arrangement to offset a variety of forces. The dam's self-weight, the hydrostatic pressure from the impounded water, the uplift pressure from seepage beneath the foundation, and occasionally dynamic factors like seismic loads or ice and wind pressure are some of these forces. To guarantee the dam's long-term functionality, safety, and structural integrity, an accurate assessment of these forces is essential.

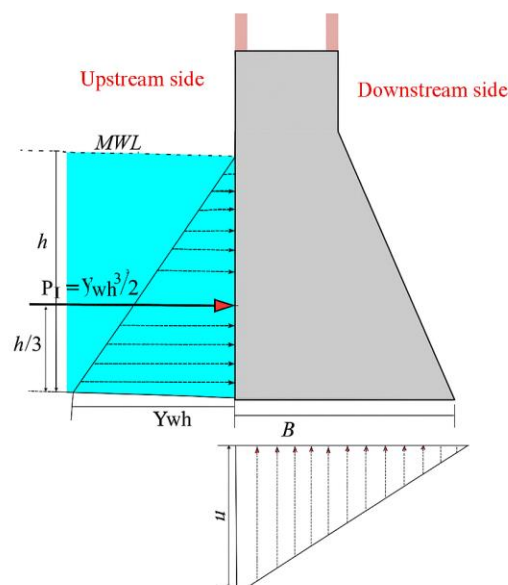


Figure 8 : Forces Acting on a Dam

Force	Equation	Magnitude
Hydrostatic Pressure, P1	$\frac{1}{2} \cdot \gamma_w \cdot h^2$	30537.1 kN/m
Uplift Pressure, P2	$\frac{1}{2} \cdot \gamma_w \cdot h \cdot B_{low}$	21497.3 kN/m
Weight of Dam 1, W1	$\gamma_c \cdot h \cdot T$	840 kN/m
Weight of Dam 2, W2	$\frac{1}{2} \cdot \gamma_c \cdot (h - 5) \cdot (B_{low} - T)$	42192.1 kN/m

Table 20 : Forces Acting on a Dam

6.1.1 SLIDING FAILURE

In gravity dams, sliding failure is a type of structural instability that happens when the horizontal driving forces acting on the dam are greater than the resisting forces that have formed along its base or possible failure planes in the foundation. The weight of the dam itself and the friction between the foundation material and the dam base are the sources of these resisting forces. The F.S.S can be determined by,

$$F.S.S. = \frac{\mu \sum V}{\sum H}$$

μ is the coefficient of friction between the dam base and foundation, which assumes a value of 0.70.

$$FSS = \frac{\mu \cdot (W_1 + W_2 - P_2)}{P_1}$$

The factor of safety against sliding was determined to be 1.1 which is greater than 1.0, hence the dam is safe against sliding.

6.1.2 OVERTURNING/ROTATION FAILURE

The term "overturning" describes a gravity dam's possible rotational instability, which usually happens around its downstream toe. This occurs when the moment of resistance offered by the dam's self-weight and vertical loads is less than the moment created by horizontal forces (such as hydrostatic pressure). If the resulting force is outside the middle third of the dam base, the dam may rotate forward around the toe. This may result in foundation distress, a loss of structural integrity, or even a catastrophic failure. The factor of safety against overturning (F.S.O.) often exhibits a range of values between 1.5 and 2.

$$FSS = \frac{\sum RM}{\sum OM}$$

RM = The total moment that prevents the dam from toppling over (The weight of the dam and its parts acting at a specific distance from the toe (downstream edge) of the dam is the main cause of these moments. The structure is stabilized and kept from rotating by the resisting moment).

OM = The total moment that tends to cause the dam to overturn about its toe (These are mostly brought on by horizontal forces like the upstream face's hydrostatic water pressure, pressure of uplift at the base, seismic or earthquake forces (if considered))

The factor of safety against overturning was determined to be 1.6 which is between the given range, hence the dam is safe against overturning. (Appendix F)

6.2 EXCAVATION

One of the most important stages of building a dam is excavating the foundation. In order to create a solid and stable foundation for the dam structure, soil and rock layers must be removed. The overall stability and long-term functionality of the dam depend on this procedure. The ability of the dam to resist external forces like water pressure, earthquakes, and environmental loading is directly impacted by the success of this phase. Silt is one of the materials that is frequently found during excavation. River valleys and alluvial deposits are common places to find silt, a fine-grained sediment that ranges in size from sand to clay. Because of its softness and water-retention properties, silt has historically been viewed as less ideal, but in some situations, it can provide significant benefits. Specifically, silt shows good adaptability to the topography of the excavation, forming a continuous and smooth surface that can sustain uniform load distribution from the dam structure. The site's silt can lessen the need for imported materials if it is used properly. This minimizes construction waste, lowers expenses, and increases resource efficiency. It has also been discovered that silt has advantageous seismic behavior. Its damping qualities aid in absorbing vibrations and lowering the possibility of foundation instability during earthquakes in areas where seismic activity is a concern.

But when excavation reaches areas with hard or diverse rock formations, problems occur. These could make the excavation slopes unstable, raising the possibility of collapse or slippage. As a result, a thorough evaluation of the sand-silt-clay composition's mechanical behavior is required. These assessments also aid in guaranteeing that the foundation material is appropriate for the environmental conditions and design loads of the dam.

In conclusion, the excavation procedure is essential to the structural safety of the dam in addition to preparing the physical foundation for its construction. The performance of the foundation can be greatly

impacted by the proper handling and assessment of materials like silt, particularly when paired with a scientific evaluation of slope stability and seismic resistance.

6.3 FOUNDATION DESIGN

A dam's foundation is an essential part that has a direct impact on its performance, stability, and longevity. Numerous important factors, including geological conditions, structural requirements, and practical feasibility during construction, influence the choice of foundation type. Among these, whether a solid rock foundation, an earth foundation, or a mixed (diverse) foundation is best suited for the site depends largely on the geological makeup of the dam site.

A thorough engineering geological survey is necessary in order to make an informed choice. This entails assessing the characteristics of rock formations, the materials' accessibility, and possible hazards like groundwater uplift pressure, filtration, and erosion. Knowing these factors guarantees that the foundation can support the dam's structural weight and endure environmental stresses over time.

Because of its high load-bearing capacity and resistance to deformation, a solid rock foundation is given priority in this project. Solid rock foundations, like granite, are especially good at stopping seepage and structural shifting. In seismic zones, where foundations must be able to absorb and disperse earthquake forces without endangering the structure, this is particularly crucial. Furthermore, strong rock foundations support the dam's long-term stability and safety. Utilizing the site's excavated rock material allows for resource-efficient construction and has additional economic and environmental advantages.

6.4 DAM SEEPAGE

The movement or infiltration of water through the dam's body, foundation, or surrounding abutments is referred to as dam seepage. The structural safety, longevity, and functionality of the dam are seriously threatened by uncontrolled or excessive seepage, even though some seepage is normal and even expected in hydraulic structures. The possibility of foundation erosion is among the most important issues associated with seepage. Water seeping into the rock or foundation soil beneath a dam can erode fine particles over time, causing voids and weakening the structural base. It can also cause piping, a phenomenon in which water carries soil materials through the foundation, forming continuous channels that can eventually cause the dam to fail catastrophically. Seepage may also have the following additional effects: reduction in water storage capacity due to reservoir water loss, instability of the dam's structure, particularly when the water level varies, elevated uplift pressure, especially in gravity or concrete dams

where water pushes upward at the foundation-dam interface, The dam's surrounding soil is deteriorating, leaving it vulnerable to settlement or landslides, development of foundation joints or cracks, which increases seepage and structural susceptibility.

Before construction, thorough geotechnical studies were conducted to evaluate and manage seepage. Understanding the permeability, stratification, and jointing of the underlying soil and rock layers is aided by these investigations. These results are used to determine the best seepage mitigation techniques.

Seepage is managed using a variety of engineering solutions:

Grouting: It is the process of reducing permeability in rock joints and cavities by injecting cementitious or chemical materials. Grouting was chosen as the main technique for this project because it effectively seals voids and lowers hydraulic conductivity.

Cutoff walls: Permeable layers are used to create vertical barriers that stop water flow. Clay blankets are horizontal, impermeable layers that are positioned upstream to reduce infiltration.

Drainage systems: Toe drains, relief wells, or filters that gather and securely release seeped water to lower pore pressure.

These actions not only prevent seepage but also shield the dam from uplift, preserve slope stability, and lengthen the structure's lifespan.

6.5 CONSTRUCTION AND MAINTENANCE

A dam's safe operation, long-term durability, and functionality depend on a number of planned and interconnected phases during construction and maintenance. Every phase, from planning to continuing upkeep, is directed by engineering principles, environmental factors, and real-world difficulties unique to the location and project objectives.

It starts with a thorough planning stage that includes carrying out in-depth feasibility studies. These studies aid in determining the necessity of the dam, evaluating its technical and financial viability, and assessing its possible effects on the local environment and communities. Numerous site-specific data are gathered to support this assessment, such as hydrological flow patterns, geological profiles, rainfall records, catchment characteristics, and topography. This phase's goal is to gain a comprehensive understanding of the site's conditions in order to inform decisions that will influence the design and construction methodology.

During the design stage, engineers create a comprehensive blueprint for the dam structure by interpreting the results of the planning phase. This involves choosing the best kind of dam based on hydrological and

geological considerations, such as an embankment, arch, or concrete gravity. The dam's size, reservoir capacity, spillway size, and energy dissipation characteristics are also determined by the design. To make sure the dam can sustain loads from water pressure, earthquakes, wind, and uplift forces, crucial calculations are performed. Along with environmental preservation and downstream population safety, the design process incorporates sustainability and regulatory requirements.

The project enters the construction phase after the design is completed. This entails preparing the foundation, clearing the site, and mobilizing resources. This stage may entail substantial excavation, rock grouting, cutoff wall installation, embankment filling, and concrete placement, contingent on the type of dam. To guarantee operational effectiveness and safety, structural components such as gates, intake towers, spillways, and monitoring systems are put in place.

Modern tools and methods are used during construction to guarantee accuracy and excellence. To ensure that the construction satisfies the necessary engineering standards and codes, strict quality control procedures are put in place, such as routine material testing and site inspections. Successful construction depends on effectively managing the construction process, resource allocation, schedules, and unforeseen technical difficulties. During all phases, the safety of workers will be ensured.

The dam moves into the operation and maintenance phase following the completion of physical construction and commissioning. At this point, the dam starts to perform the tasks for which it was designed, including storing water, managing floods, producing hydroelectric power, or providing irrigation. This phase does not, however, mark the conclusion of engineering involvement. To maintain the dam's integrity and functionality, routine maintenance and inspections are essential.

Structural examinations, debris removal, reservoir desilting, mechanical component cleaning and lubrication, and watch for seepage, deformation, or distress indicators are typical maintenance tasks. Engineers can evaluate performance over time with the use of specialized instruments installed inside the dam, which provide real-time data on stresses, displacements, and water levels. The early possible failures can be spotted and take prompt action by regularly inspecting and maintaining the dam.

6.6 MONITORING AND CONTROL DEVICES

Strong monitoring and control systems must be put in place for dams to be safe and reliable. Advanced systems that facilitate early warning, ongoing observation, and efficient response to possible structural or functional issues must be incorporated due to the growing importance of dam safety. Because embankment dams are susceptible to seepage, deformation, and changes in internal pressure, it is imperative that

comprehensive monitoring technologies be integrated.

Key parameters that are monitored include:

- i. **Seepage monitoring:** Unchecked seepage can cause internal erosion, structural deterioration, and ultimately dam failure. Prompt remediation and dam stability are guaranteed by early detection.
- ii. **Deformation monitoring:** It helps find early indicators of possible instability or structural damage by detecting any changes or displacements in the dam's structure.
- iii. **Leak Detection:** In order to reduce water loss and stop erosion or flooding in the surrounding areas, leak detection is essential for locating and identifying water escape points.
- iv. **Pore Pressure Monitoring:** Assists in identifying variations in internal water pressure, which could be a sign of internal degradation brought on by stress changes or the advancement of seepage.
- v. **Water Level Monitoring:** Constant monitoring of reservoir levels helps determine storage capacity, seepage patterns, and emergency readiness. It also acts as a performance indicator.

An advanced dam monitoring system must incorporate a network of sensors, data collection devices, and real-time analytics platforms to effectively monitor these parameters. In embankment dams, technologies like temperature field analysis and seepage-heat monitoring are especially useful for locating possible failure zones and seepage routes.

Furthermore, environmental effects should be included in monitoring plans in addition to structural and hydraulic ones. To find anomalies like erosion, leaks, or mechanical wear, routine inspections and system upgrades are essential. Dam authorities can guarantee the structure's long-term stability, safety, and performance while lowering the likelihood of unplanned failures by using a proactive and comprehensive monitoring approach.

7.0 DRAWING

7.1 STORAGE LEVEL

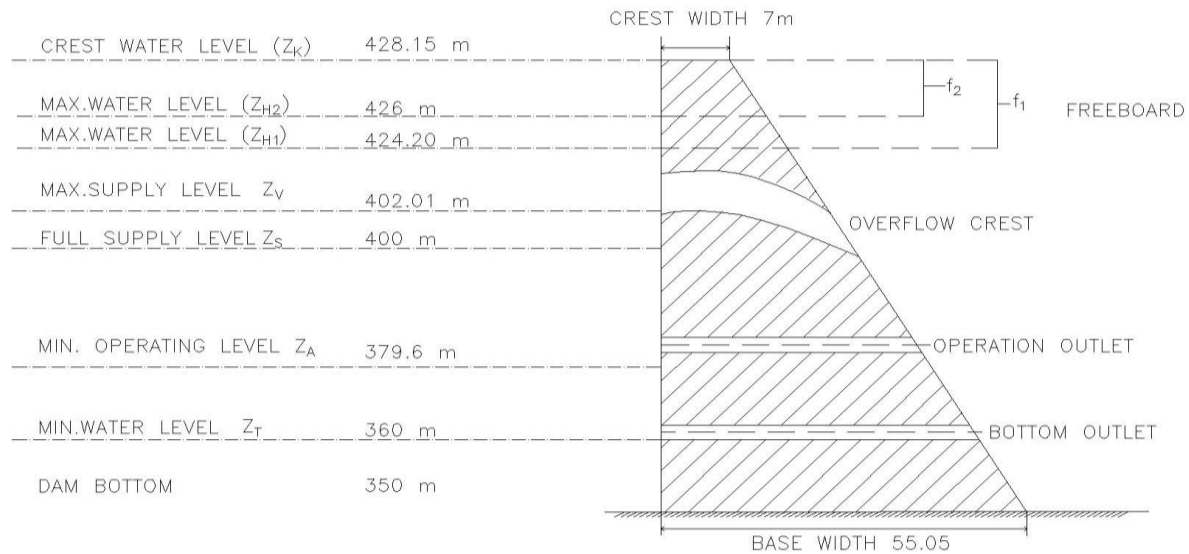


Figure 9 Storage Level Of Reservoir Designed, (Drawn Using AutoCAD)

7.2 WATER LEVEL

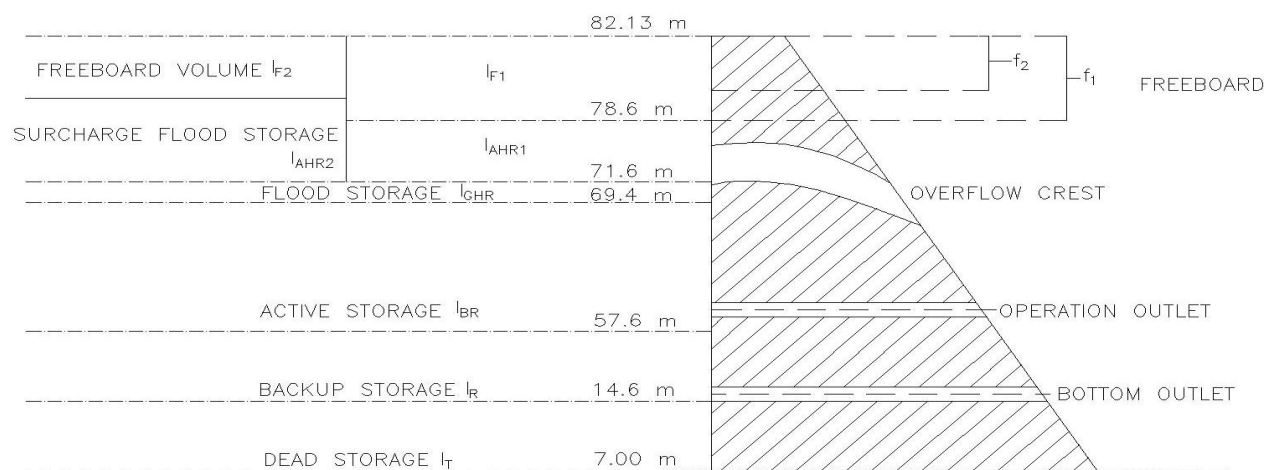


Figure 10 Water Level Of The Reservoir (Designed Using AUTOCAD)

7.3 ENERGY DISSIPATION

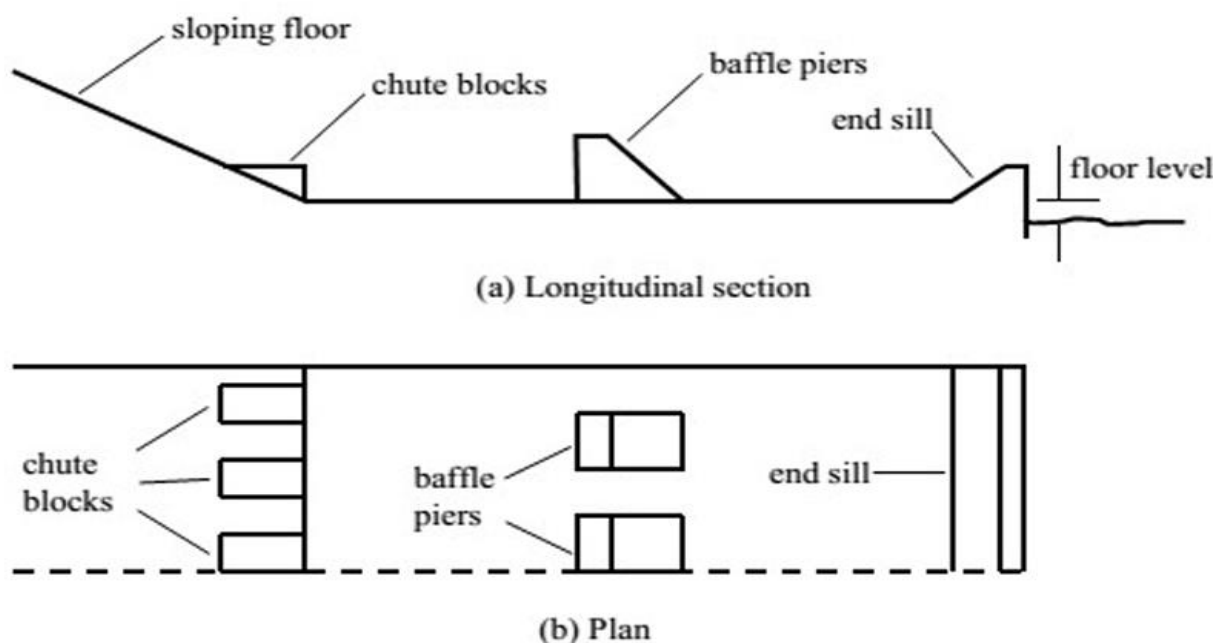


Figure 11: Energy Dissipation [13]

APPENDIX A

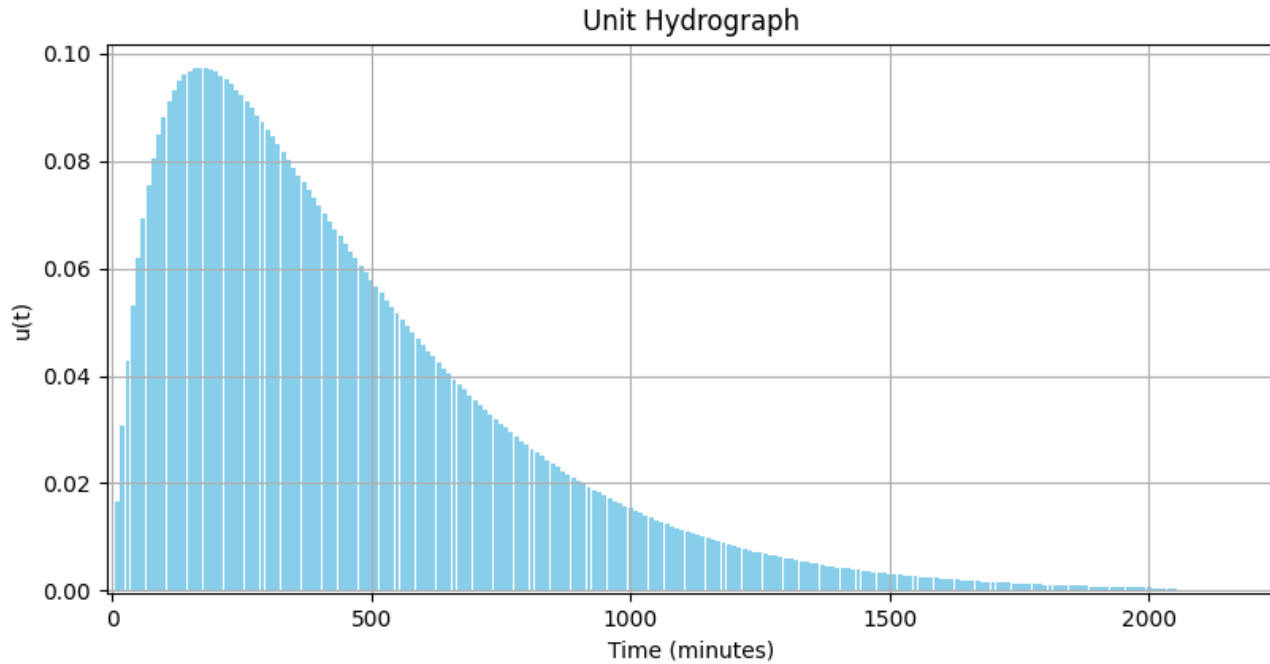
ACTIVE STORAGE CALCULATION

Monthly drinking water demand.							
Month	Rqd Volume	Rqd Vol m ³	No: of Days	Available Supply	Net(m ³)	Cum Storage	in million
Jan	0,96	960000	31	803520	-156480	156480	0,15648
Feb	0,88	880000	28	725760	-154240	310720	0,31072
Mar	1,36	1360000	31	803520	-556480	867200	0,8672
Apr	1,84	1840000	30	777600	-1062400	1929600	1,9296
Mai	2,16	2160000	31	803520	-1356480	3286080	3,28608
Jun	2,64	2640000	30	777600	-1862400	5148480	5,14848
Jul	4,08	4080000	31	803520	-3276480	8424960	8,42496
Aug	2,24	2240000	30	777600	-1462400	9887360	9,88736
Sep	2,56	2560000	31	803520	-1756480	11643840	11,64384
Oct	2,00	2000000	31	803520	-1196480	12840320	12,84032
Nov	1,52	1520000	30	777600	-742400	13582720	13,58272
Dec	1,12	1120000	31	803520	-316480	13899200	13,8992

i	$Q_{in,i}$	Q_{in} Adjusted	$Q_{out,i}$	$Q_{out,ecological}$	$Q_{out,Total}$	$Q_{in,i} - Q_{out,i}$	SD_i	HP_i	LP_i	C_i
1	3,85	3,40	0,96	0,40	1,36	2,04	2,04			
2	3,81	3,37	0,88	0,36	1,24	2,13	4,17			
3	3,06	2,70	1,36	0,40	1,76	0,94	5,11			
4	3,74	3,31	1,84	0,78	2,62	0,69	5,80	5,8		
5	3,07	2,71	2,16	0,80	2,96	-0,25	5,55			
6	3,11	2,75	2,64	0,78	3,42	-0,67	4,88			
7	3,40	3,00	4,08	0,80	4,88	-1,88	3,00			
8	2,99	2,64	2,24	0,80	3,04	-0,40	2,60			
9	2,85	2,51	2,56	0,78	3,34	-0,83	1,77		1,77	4,03
10	2,84	2,51	2,00	0,40	2,40	0,11	1,88			
11	2,65	2,34	1,52	0,39	1,91	0,43	2,31			
12	4,07	3,60	1,12	0,40	1,52	2,08	4,39			
13	1,22	1,08	0,96	0,40	1,36	-0,28	4,11			
14	2,30	2,03	0,88	0,36	1,24	0,79	4,90			
15	2,74	2,42	1,36	0,40	1,76	0,66	5,56			
16	3,51	3,10	1,84	0,78	2,62	0,48	6,04			
17	7,80	6,89	2,16	0,80	2,96	3,93	9,97			
18	3,90	3,44	2,64	0,78	3,42	0,02	10,00	10,00		
19	2,74	2,42	4,08	0,80	4,88	-2,46	7,54			
20	3,94	3,48	2,24	0,80	3,04	0,44	7,98			
21	2,74	2,42	2,56	0,78	3,34	-0,92	7,06		7,06	2,94
22	2,97	2,63	2,00	0,40	2,40	0,23	7,29			

APPENDIX B

UNIT HYDROGRAPH



PYTHON CODE

```
import numpy as np
import matplotlib.pyplot as plt
Constants (use the provided values)
beta = 0.1483 K1 = 1.5874 # 1/h K2 = 4.2117 # 1/h
Time in minutes: from 10 to 2000 in steps of 10
t_minutes = np.arange(10, 1451, 10)
Convert minutes to hours
t_hours = t_minutes / 60
Unit hydrograph components
u1 = (t_hours / K12) * np.exp(-t_hours / K1) u2 = (t_hours / K22) * np.exp(-t_hours / K2) u
= beta * u1 + (1 - beta) * u2
Plot
plt.figure(figsize=(12, 6)) plt.bar(t_minutes, u, width=9, color='skyblue')
plt.xlabel("Time (minutes)") plt.ylabel("u(t)") plt.title("Unit Hydrograph (10-minute
intervals up to 1450 minutes)") plt.grid(True) plt.tight_layout() plt.show()
```

APPENDIX C

CATCHMENT PARAMETERS

1. Catchment area (A): $72.6 \text{ km}^2 = 72,600,000 \text{ m}^2$
2. Baseflow: $0.0976 \text{ m}^3/\text{s}$
3. Outflow capacity: $17.08 \text{ m}^3/\text{s}$
4. Durations analyzed: 6h, 9h, 12h, 18h, 24h, 48h, 72h
5. Rainfall distributions: Uniform, Advanced, Intermediate, Delayed
6. Unit hydrograph: 6 blocks per storm event with assigned multipliers per pattern

CALCULATION STEPS

Effective Rainfall (in mm) for each duration and pattern was taken from the KOSTRA-DWD-2020 data, adjusted for initial and constant losses.

Rainfall was split into 6 equal time blocks for each pattern.

Rainfall volume per block was calculated using:

$$V_{block} = \left(\frac{r_{block}}{1000} \right) \times A$$

Flow response (Q_i) for each block was estimated using a unit hydrograph:

$$Q_i = \frac{V_{block} \times UH \text{ multiplier}}{block \text{ duration (s)}}$$

If, $Q_i > Q_{out}$ the excess flow was calculated and converted into flood volume:

$$V_{flood} = (Q_i - Q_{out}) \times duration (s)$$

Total flood volume is the sum of flood volumes for all 6 blocks.

Flood Volume Calculation for 6h Uniform Pattern

Rainfall Volume per Block

$$V = \frac{4,149}{1000} \times 7,260,000 = 300,189.9 \text{ m}^3$$

Flow per Block

Block	UH Multiplier	Q_i
1	0.6	50.20
2	1.2	100.40
3	1.8	150.59
4	1.6	133.86
5	1.0	83.66
6	0.4	33.47

Excess Flow and Flood Volume Per Block

Block	Q_{excess}	Volume (m ³)
1	33.12	119224.97
2	83.32	299937.93
3	133.51	480650.90
4	116.78	420413.28
5	66.58	239700.28
6	16.39	58987.31

Total Flood Volume = 1618914.06 m³

APPENDIX D

STORAGE CURVE

The water level and storage volume curve is obtained from the below table

Volume [10^6 m^3]	Water Level [m]
0	350
7.02641	360.2125
9.64856	379.6823
19.23077	393.8757
38.46154	400.9424
41.46154	416.2168
44.87179	425.7197
51.28205	434.7197
57.69231	441.9343
64.10256	448.8313
70.51282	455.0294
79.92308	460.5995
83.33333	470.1036
89.74359	474.1463
96.15385	477.7793
102.5641	481.0422
108.9744	483.9782
115.3846	486.615
121.2051	488.9845
128.2051	491.114
134.6154	493.0277
141.2056	494.7475
147.4359	496.293
153.8642	497.6819
160.2564	498.9301

APPENDIX E

FREE BOARD CALCULATION

To determine the required freeboard, a modular approach is applied in accordance with DIN 19700 and supported by methodologies outlined in DWA-M-552 and DVWK-M-246. This method consider the cumulative effects of:

- (1) Wind-induced waves run up on the reservoir surface
- (2) Wind setup (quasi-static rise in water surface due to wind pressure)
- (3) Ice jam formation
- (4) A safety surcharge (in more conservative scenarios)

Two design scenarios are evaluated based on the return period of the wind event during the flood.

Case 1: Wind return period $T_{\text{Wind}} = 25$ years

Case 2: Wind return period $T_{\text{Wind}} < 25$ years

Each case is calculated as:

Case 1:

$$f_1 = h_{Au} + h_{\omega_i} + h_{E_i}$$

Where:

$$h_{Au} = 1.2 \text{ m (Wave run-up height)}$$

$$h_{\omega_i} = 0.05 \text{ (Wind setup)}$$

$$h_{E_i} = 0.5 \text{ (Ice jam height)}$$

$$f_1 = 1.2 + 0.05 + 0.5 = 1.75 \text{ m}$$

Case 2:

$$f_2 = h_{Au} + h_{\omega_i} + h_{E_i} + h_{s_i}$$

Where:

$$h_{s_i} = 0.4 \text{ (Safety surcharge)}$$

$$f_2 = 1.2 + 0.05 + 0.5 + 0.4 = 2.15 \text{ m}$$

Given the strategic importance and safety classification of the structure, and considering that precise local wind recurrence data may not be available with high certainty, the Conservative **Case 2** is adopted for

design:

Adopted Freeboard = 2.15 m

APPENDIX F

FORCES ACTING ON THE DAM

$$h = 78.15 \text{ m}$$

$$B_{low} = 55.05 \text{ m}$$

$$T = 7 \text{ m}$$

$$\gamma_w = 10 \frac{\text{kN}}{\text{m}^3}$$

$$\gamma_c = 24 \frac{\text{kN}}{\text{m}^3}$$

$$\mu = 0.7$$

$$\text{Hydrostatic Force (P1)} = 0.5 \cdot 10 \cdot 78.15^2 = 30537.1 \frac{\text{kN}}{\text{m}}$$

$$\text{Uplift Force (P2)} = 0.5 \cdot 10 \cdot 78.15 \cdot 55.05 = 21497.3 \text{ kN}$$

$$\text{Weight of Dam} = W1 + W2$$

$$W1 = 24 \cdot 5 \cdot 7 = 840 \text{ kN/m}$$

$$W2 = 0.5 \cdot 24 \cdot 73.15 \cdot (55.05 - 7) = 42192.1 \text{ kN/m}$$

$$\text{Total downward Force} = W1 + W2 = 43032 \text{ kN}$$

FACTOR OF SAFETY

$$FOS_{sliding} = \frac{\mu(\sum W - P_2)}{P_1}$$

$$= \frac{0.7(43032 - 21497)}{30537} = 1.1$$

Moments resisting overturn (MR) :

$$MR = W_1 \cdot \frac{B}{2} \cdot W_2 \cdot \frac{B}{3}$$

$$\text{Moment of } W_1 = 840 \cdot \frac{55.05}{2} = 23121 \text{ kNm}$$

$$\text{Moment of } W_2 = 42192 \cdot \frac{55.05}{3} = 774067 \text{ kNm}$$

$$MR = 23121 + 774067$$

$$MR = 797188 \text{ kNm}$$

Moments causing overturn (MO):

Horizontal Water force acts at $\frac{h}{3}$ from base

$$MO = P_1 \cdot \frac{h}{3} = 30537 \cdot \frac{78.15}{3} = 796478 \text{ kNm}$$

$$FOS_{\text{Overturning}} = \frac{MR}{MO}$$

$$FOS_{\text{Overturning}} \approx 1.6$$

PROJECT MANAGEMENT STATEMENT 1

The phrase “Fountain of Youth” is a legendary concept from folklore and mythology. It refers to a mythical spring or fountain that supposedly restores the youth and vitality of anyone who drinks from it or bathes in its waters. The idea of a rejuvenating fountain appears in ancient mythologies, including Greek, Arabic, and Asian traditions. In the Age of Exploration, the legend gained popularity when Spanish explorer Juan Ponce de León was said to be searching for the Fountain of Youth in Florida in the early 1500.

We choose this name as the reflection of our task, to design a drinking water reservoir which aims to bring renewal and resilience to the region Thüringen, which supports the communities and strengthens the environment.

Modern Relevance - Water is one of the most important or the most important resources for life on earth. As the time passed it became apparent that drinking water became a luxury in many parts of the world and questioned the existence of life on earth in the coming ages. So as students of sustainable hydrology, it is our duty to do everything in our power and knowledge to conserve it for the future. The modern relevance of the Fountain of Youth lies in how it symbolizes our ongoing quest to slow down, reverse, or avoid extinction — a pursuit that now blends knowledge with science and sustainability.

Team members

Akshay Maru Veettil (03796381)

Arya Aji (03796925)

Krishna Vinod (03796402)

Sarath Raj Moolayil Veettil (03796398)

Role and task of members

	TASK	RESPONSIBLE MEMBER
1	Data Collection and Site Analysis	Akshay, Arya Aji
2	Drinking Water Demand Analysis	Arya Aji, Krishna
3	Flood Estimation and Hydrological Calculations	Everyone
4	Definition of Boundaries	Sarath, Krishna
5	Dam Type Selection	Sarath, Arya
6	Volume and Reservoir Area Estimation	Krishna, Akshay
7	Selection of Hydraulic Structures	Sarath, Krishna
8	Infrastructure Planning	Sarath, Akshay
9	Report Structuring and Formatting	Everyone

Communication contact person

Sarath Raj Moolayil Veettil (03796398) - go73zob@mytum.de

Timeline and dependencies

Task	Timeline	Description
Data Collection & Site Analysis	13–17 May	Collect GIS and hydrological data, assess project location and soil details should be collected
Drinking Water Demand analysis	20–27 May	Calculate water needs
Flood Estimation	23 May – 1 June	Estimate design floods (BHQs, RQ)
Project Management Statement I	30 May	Submit the PMS I according to the criteria

Submission		
Boundary & Volume Planning	5 – 14 June	Define reservoir boundaries and estimate required storage volume and area.
Dam type finalizing	16–28 June	Select dam type and plan spillway and outlet structures
Infrastructure Planning & Drafting	1–12 July	Plan road relocations, recreation zones
Final Editing & Project Management Statement correcting	15–24 July	Final edits and review; submit Project Management Statement II
Project Management Statement II submission	25 July	Submit the PMS II according to the criteria
Draft Report Writing	1 July	Start the report work
Final editing of Report	25–28 July	Compile final report with figures, references, and appendices
Final Report Submission	By 29 July	Submit report on Moodle

PROJECT MANAGEMENT STATEMENT 2

Our team worked together throughout the project to divide and complete the different technical tasks related to the hydraulic dam design. With the explicit goal of ensuring balanced participation throughout the project, the initial task distribution was determined by each member's technical interests and strengths. However, because some team members were unavailable at various points during the project timeline, we ran into some difficulties. This made it challenging to adhere strictly to our original task allocation.

We reorganized the tasks among available members and conducted internal discussions to make sure the project moved forward effectively despite these challenges. This redistribution made it possible for us to promptly address the outstanding portions and guaranteed that each significant project component received concentrated attention. We also divided each task into subtasks to reduce the error and stress on individuals.

Updated Work Distribution Approach

	Task	Responsible Member
1	Data Collection	Arya, Akshay
2	Reservoir Capacity Planning	Krishna
3	Dam Type Selection	Krishna
4	Discharge and Storage Calculation	Krishna
5	Flood Storage Calculation	Krishna
6	KOSTRA data collection and Time Distribution	Akshay
7	Surcharge Flood Storage	Sarath
8	Water Level, Volume, and Freeboard Calculation	Arya
9	Dam Geometry	Arya, Akshay
10	Operational Facilities	Sarath
11	Structure Design	Sarath
12	Stability Analysis	Krishna
13	Monitoring and Controlling Devices	Krishna
14	Drawing	Akshay, Arya

We reorganized the remaining tasks by combining related technical components into groups and allocating them to members who were available after reevaluating the workload. This reduced delays and guaranteed more seamless coordination. Additionally, the team collaborated on overlapping sections where accuracy required data sharing and cross-verification. With flexibility granted to members to finish their sections within reasonable timeframes, the redistribution enabled us to maintain a balanced effort among members towards the report's completion.

Even though we changed the responsible person, the work was strictly followed by the time schedules. i.e., we were able to finish our tasks within the time. Now we are finalizing the figures, tables, and their labelling. Moreover, we are also checking the dimensions, notations, and reference citation styles.

25/07/2025 – Confirm that all sections are compiled together and check the dimensions and notations

26/07/2025 – Add table of contents and list of figures

27/07/2025 – Final Checking of content

28/07/2025 – Final Report Submission

By working together and redistributing the workload, we were able to finish all of the project's required components. Despite unforeseen obstacles, we maintained our flexibility and prioritized timely submission and high-quality work. This project's collaborative experience reaffirmed the value of flexibility, internal communication, and proactive problem-solving.

Our team successfully completed all tasks of the dam design project through collective effort and redistribution of tasks when some members were not available. Despite this we completed everything on time. Now we are focusing on compiling the report and finalizing it for submission.

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